8.2.6 Pin assignments

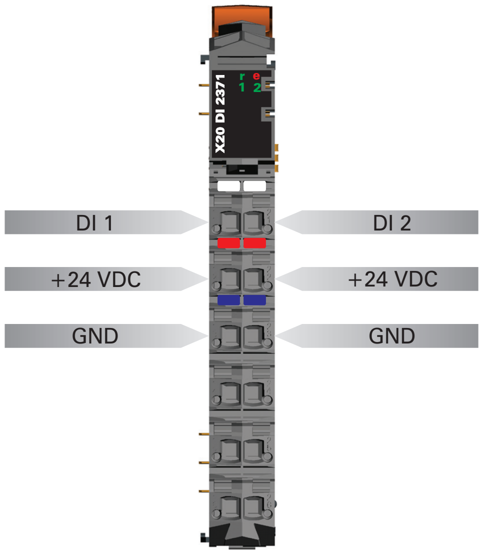


Figure 56: DI2371 - pin assignments

8.2.7 Connection example

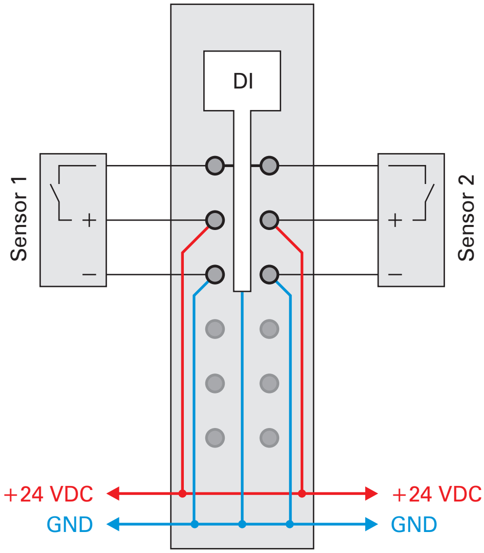


Figure 57: DI2371 - connection example

8.2.8 Input circuit diagram

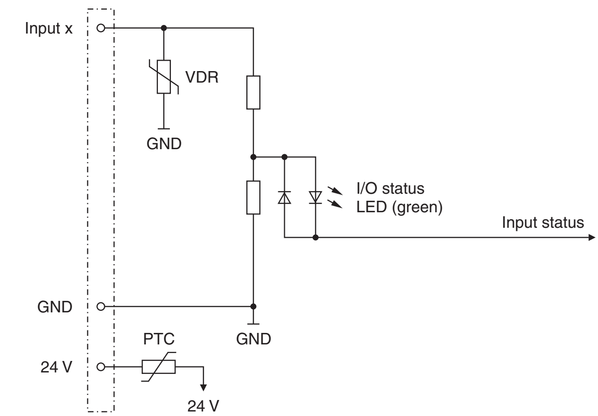


Figure 58: DI2371 - input circuit diagram

8.2.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.2.10 B&R ID code

Code for module identification ($1B8D).

8.2.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering [1]	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.2.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.3 DI2377

8.3.1 General information

The DI2377 module is equipped with two inputs for 3-wire connections. Both inputs can be configured as event counters. Gate measurement is only ever possible on one channel.

The DI2377 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 digital inputs
- Sink connection
- 3-wire connection
- 2 counter inputs with 50 kHz counter frequency
- Gate measurement
- 24 VDC and GND for sensor supply
- Software input filter can be configured for entire module

8.3.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI2377	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 2 event counters 50 kHz	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 83: DI2377 - order data

8.3.3 Technical data

Product ID	DI2377
Short description	
I/O module	Two 24 VDC digital inputs for 3-wire connections, special functions
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤10 µs Default 0 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
Additional functions for inputs	Event counting, gate measurement
Sensor supply	0.5 A total current
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.2 W Max. 0.74 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 84: DI2377 - technical data

8.3.4 Additional technical data

Product ID	DI2377
Sensor supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Sensor supply	 Max. 12.0 W 1)
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 10.5 mA
Input resistance	Typ. 2.23 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
Event counter	
Number of counters	2
Counter 1	Input 1
Counter 2	Input 2
Signal form	Square wave pulse
Input frequency	Max. 50 kHz
Count frequency	Max. 50 kHz
Counter size	16-bit

Table 85: DI2377 - additional technical data

Product ID	DI2377
Gate measurement	
Number of gate measurements	1
Gate measurement on	Input 1 or Input 2
Signal form	Square wave pulse
Evaluation	Positive edge - negative edge
Pulse length	≥20 µs
Length of pauses between pulses	≥100 µs
Internal counter frequency	48 MHz, 3 MHz, 187.5 kHz
Counter size	16-bit
General information	
B&R ID code	$1B8E

Table 85: DI2377 - additional technical data (cont.)

1) The power consumption of the sensors connected to the module may not exceed 12 W.

8.3.5 Status LEDs

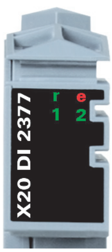
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green		Input status of the corresponding digital input

Table 86: DI2377 - status indicators

8.3.6 Pin assignments

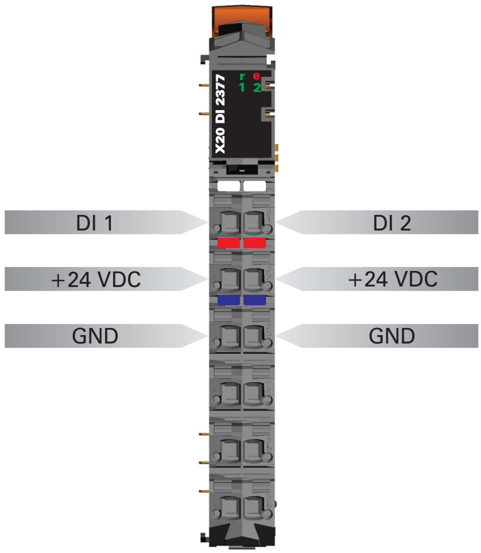


Figure 59: DI2377 - pin assignments

8.3.7 Connection example

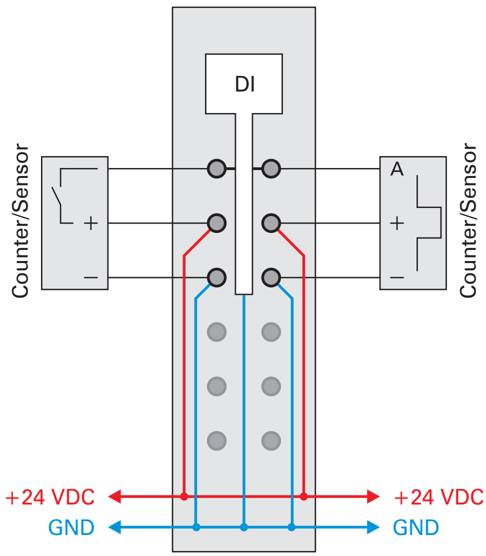


Figure 60: DI2377 - connection example

8.3.8 Input circuit diagram

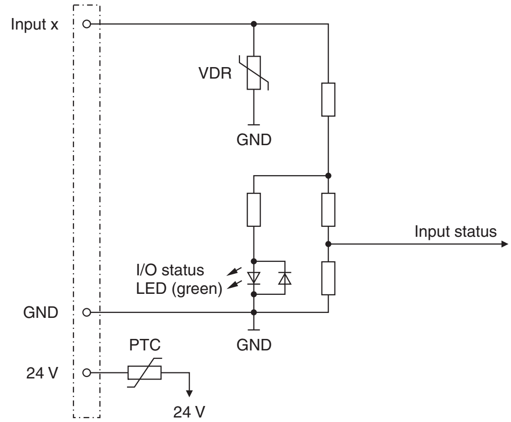


Figure 61: DI2377 - input circuit diagram

8.3.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.3.10 Counter operation

The following operation modes can be selected:

- Event counter operation
- Gate measurement

Event counter operation

The falling edges are registered on the counter input.

The counter status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Gate measurement

Information:

Only one of the counter channels at a time can be used for gate measurement.

The time of rising to falling edges for the gate input is registered using an internal frequency. The result is checked for overflow ($FFFF) and corrected with the adjustable prescaler.

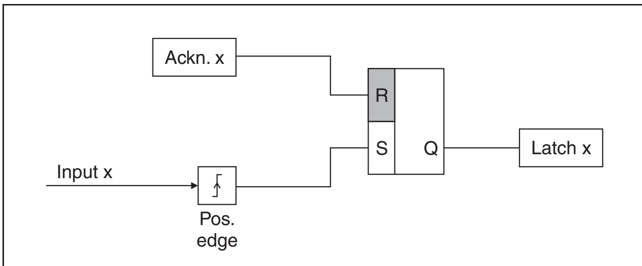
The recovery time between measurements must be >100 µs.

The measurement result is transferred with the falling edge to the result memory.

8.3.11 Input latch - positive edge

Using this function, the positive edges of the input signal can be latched with a resolution of 200 µs. With the "Acknowledge - input latch" function, the input latch is either reset or prevented from latching.

It works in the same way as a dominant reset RS flip-flop.



R	S	Q	Status
0	0	x	Do not change
0	1	1	Set
1	0	0	Reset
1	1	0	Reset

8.3.12 Acknowledge input latch

This function is used to reset the input latches channel by channel.

8.3.13 B&R ID code

Code for module identification ($1B8E).

8.3.14 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering 1)	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.3.15 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.4 DI4371

8.4.1 General information

The DI4371 module is equipped with four inputs for 3-wire connections.

- 4 digital inputs
- Sink connection
- 3-wire connection
- 24 VDC and GND for sensor supply
- Software input filter can be configured for entire module

8.4.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI4371	X20 digital input module, 4 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 87: DI4371 - order data

8.4.3 Technical data

Product ID	DI4371
Short description	
I/O module	Four 24 VDC digital inputs for 3-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
Sensor supply	0.5 A total current
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.15 W Max. 0.52 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 88: DI4371 - technical data

8.4.4 Additional technical data

Product ID	DI4371
Sensor supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Sensor supply	 Max. 12.0 W 1)
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B92

Table 89: DI4371 - additional technical data

1) The power consumption of the sensors connected to the module may not exceed 12 W.

8.4.5 Status LEDs

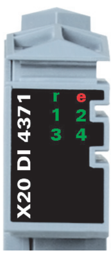
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Green		Input status of the corresponding digital input

Table 90: DI4371 - status indicators

8.4.6 Pin assignments

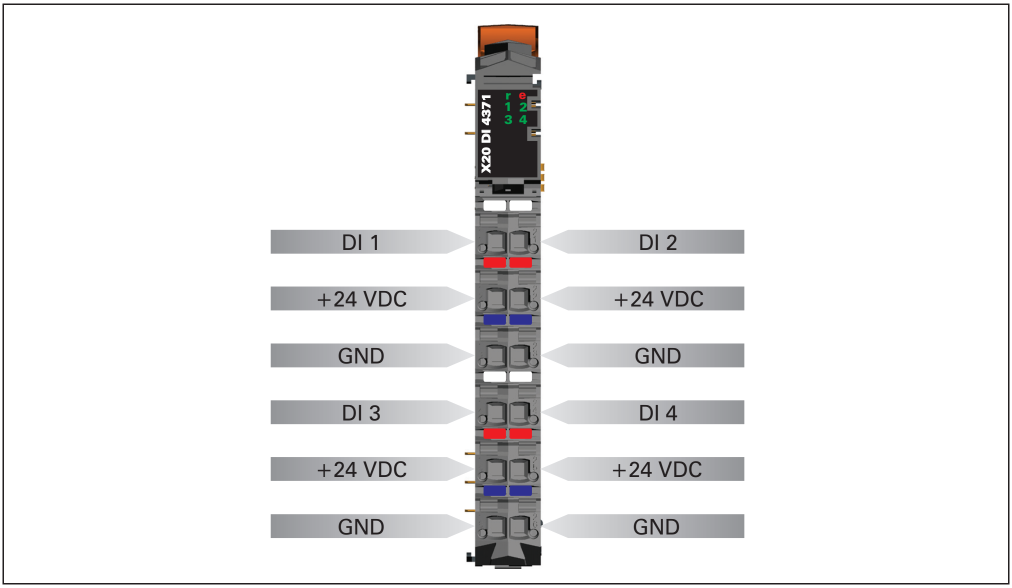


Figure 62: DI4371 - pin assignments

8.4.7 Connection example

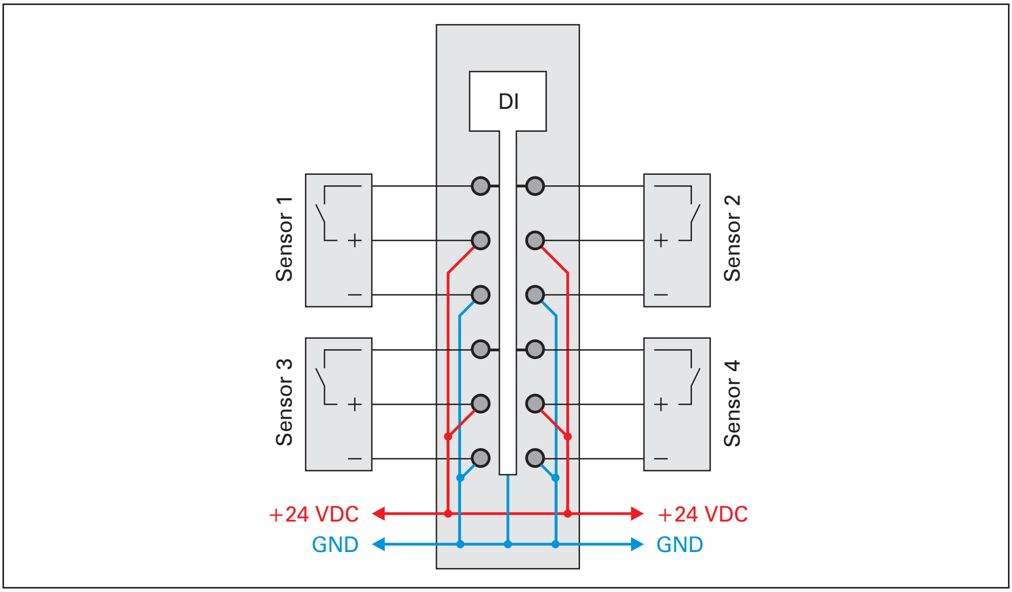


Figure 63: DI4371 - connection example

8.4.8 Input circuit diagram

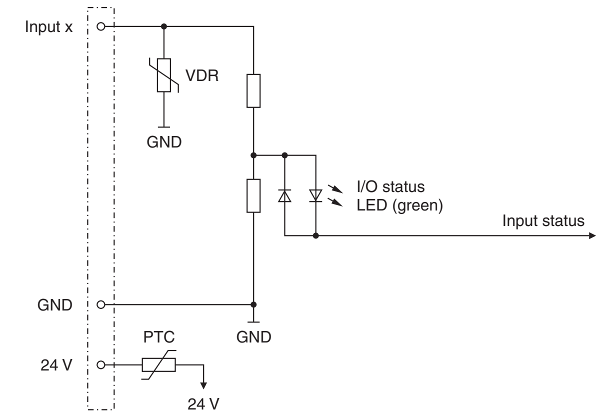


Figure 64: DI4371 - input circuit diagram

8.4.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.4.10 B&R ID code

Code for module identification ($1B92).

8.4.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering [1]	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.4.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.5 DI6371

8.5.1 General information

The DI6371 module is equipped with six inputs for 1 or 2-wire connections. The X20 standard 6-pin terminal block can be used for universal 1-line wiring. Two-line wiring can be implemented using the 12-pin terminal block. The DI6371 designed for sink input wiring.

- 6 digital inputs
- Sink connection
- 2-wire connection
- 24 VDC for sensor supply
- Software input filter can be configured for entire module
- 1-wire connection type with 6-pin terminal block

8.5.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI6371	X20 digital input module, 6 inputs, 24 VDC, sink, configurable input filter, 2-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 91: DI6371 - order data

8.5.3 Technical data

Product ID	DI6371
Short description	
I/O module	Six 24 VDC digital inputs for 1 or 2-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.16 W Max. 0.78 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 92: DI6371 - technical data

8.5.4 Additional technical data

Product ID	DI6371
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B93

Table 93: DI6371 - additional technical data

8.5.5 Status LEDs

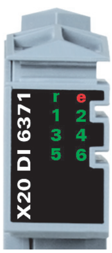
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 6	Green		Input status of the corresponding digital input

Table 94: DI6371 - status indicators

8.5.6 Pin assignments

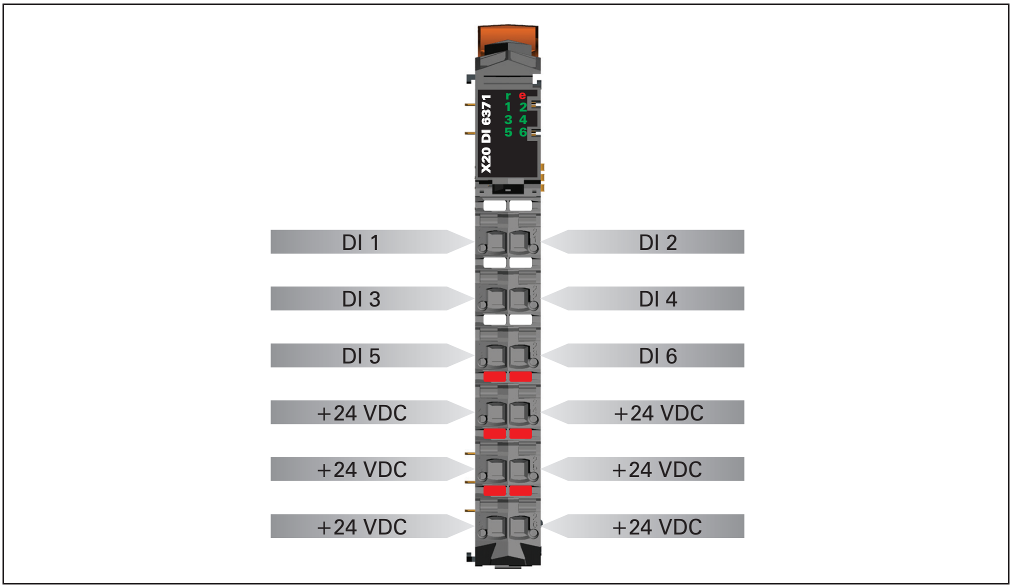


Figure 65: DI6371 - pin assignments

8.5.7 Connection example

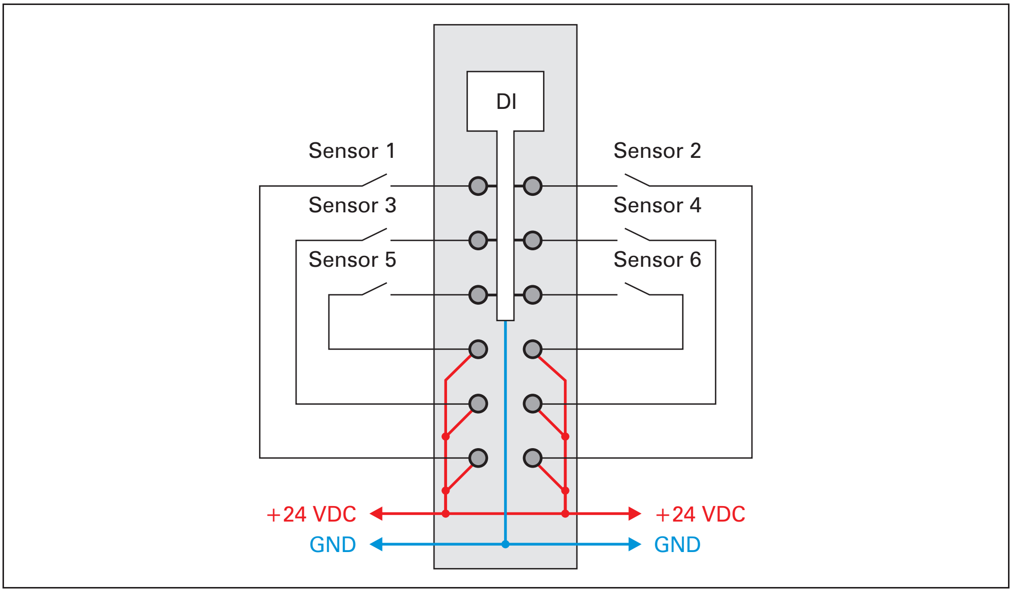


Figure 66: DI6371 - connection example

8.5.8 Input circuit diagram

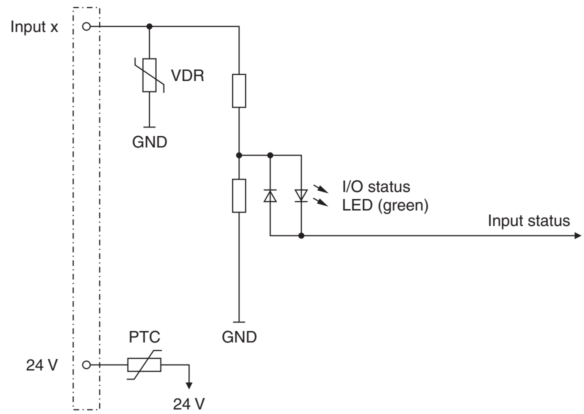


Figure 67: DI6371 - input circuit diagram

8.5.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.5.10 B&R ID code

Code for module identification ($1B93).

8.5.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering [1]	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.5.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.6 DI6372

8.6.1 General information

The DI6372 module is equipped with six inputs in 1- or 2-line connections. The X20 standard 6-pin terminal block can be used for universal 1-line wiring. Two-line wiring can be implemented using the 12-pin terminal block. The DI6372 designed for source input wiring.

- 6 digital inputs
- Source connection
- 2-wire connection
- 24 VDC for sensor supply
- Software input filter can be configured for entire module
- 1-wire connection type with 6-pin terminal block

8.6.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI6372	X20 digital input module, 6 inputs, 24 VDC, source, configurable input filter, 2-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 95: DI6372 - order data

8.6.3 Technical data

Product ID	DI6372
Short description	
I/O module	Six 24 VDC digital inputs for 1 or 2-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Source
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.16 W Max. 0.78 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 96: DI6372 - technical data

8.6.4 Additional technical data

Product ID	DI6372
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B94

Table 97: DI6372 - additional technical data

8.6.5 Status LEDs

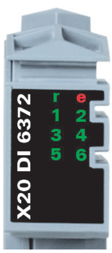
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 6	Green		Input status of the corresponding digital input

Table 98: DI6372 - status indicators

8.6.6 Pin assignments

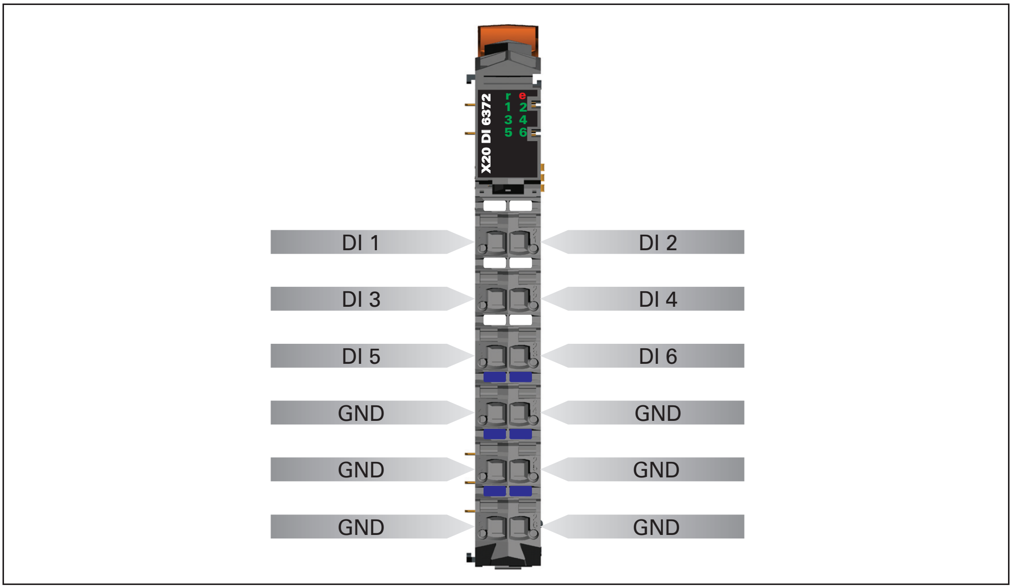


Figure 68: DI6372 - pin assignments

8.6.7 Connection example

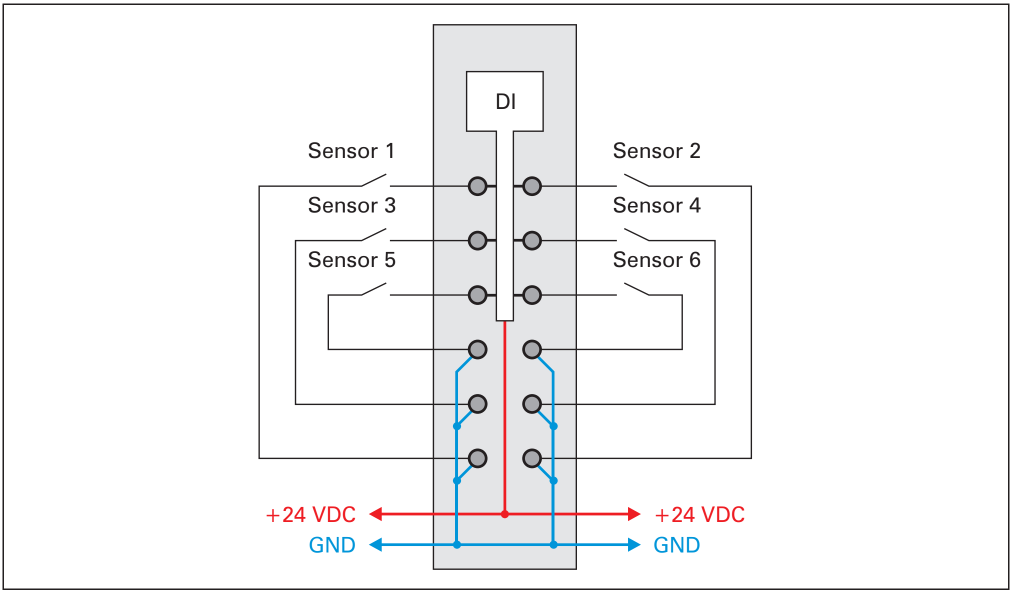


Figure 69: DI6372 - connection example

8.6.8 Input circuit diagram

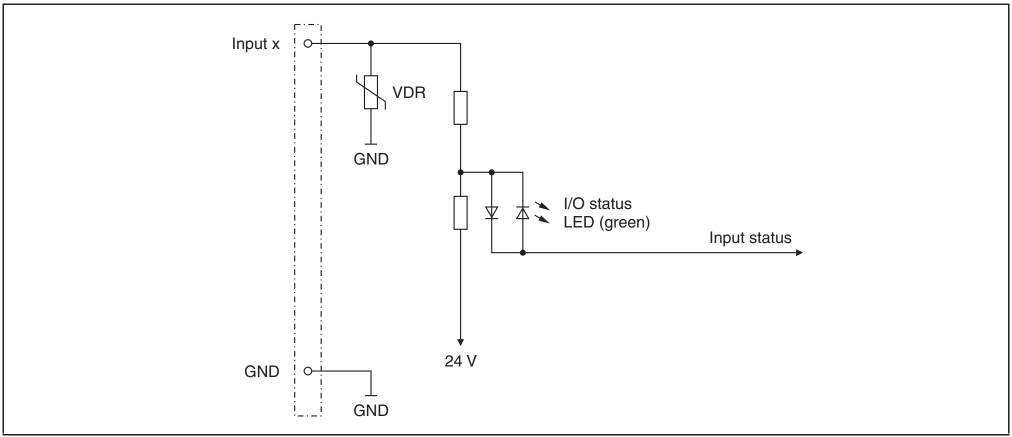


Figure 70: DI6372 - input circuit diagram

8.6.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.6.10 B&R ID code

Code for module identification ($1B94).

8.6.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering [1]	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.6.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.7 DI9371

8.7.1 General information

The DI9371 module is equipped with twelve inputs for 1-wire connections. The DI9371 designed for sink input wiring.

- 12 digital inputs
- Sink connection
- 1-wire connection
- Software input filter can be configured for entire module

8.7.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI9371	X20 digital input module, 12 inputs, 24 VDC, sink, configurable input filter, 1-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 99: DI9371 - order data

8.7.3 Technical data

Product ID	DI9371
Short description	
I/O module	Twelve 24 VDC digital inputs for 1-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal I/O external	 Typ. 0.2 W - Max. 1.56 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 100: DI9371 - technical data

8.7.4 Additional technical data

Product ID	DI9371
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B95

Table 101: DI9371 - additional technical data

8.7.5 Status LEDs

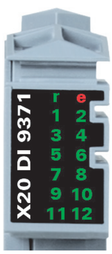
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 12	Green		Input status of the corresponding digital input

Table 102: DI9371 - status indicators

8.7.6 Pin assignments

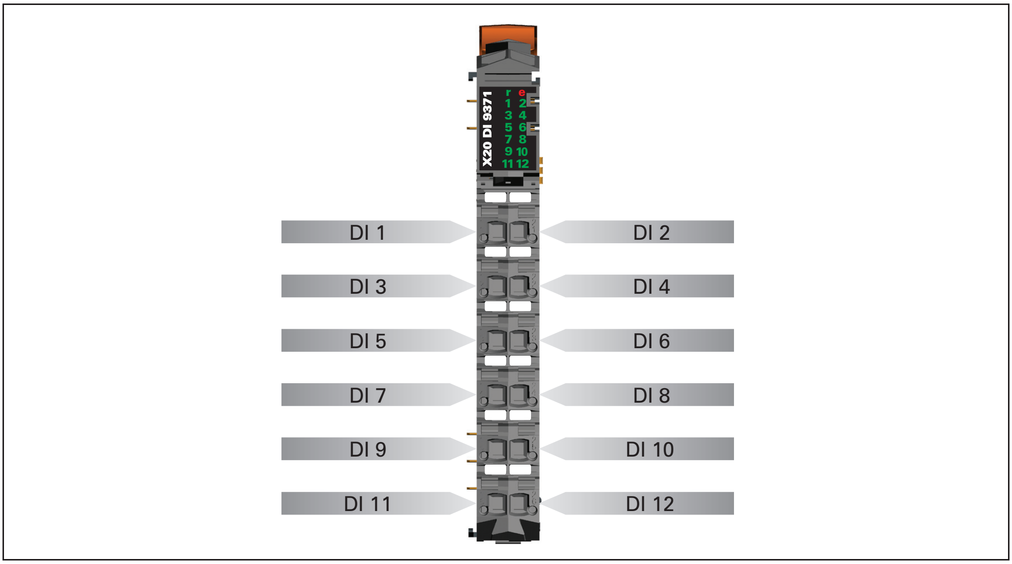


Figure 71: DI9371 - pin assignments

8.7.7 Connection example

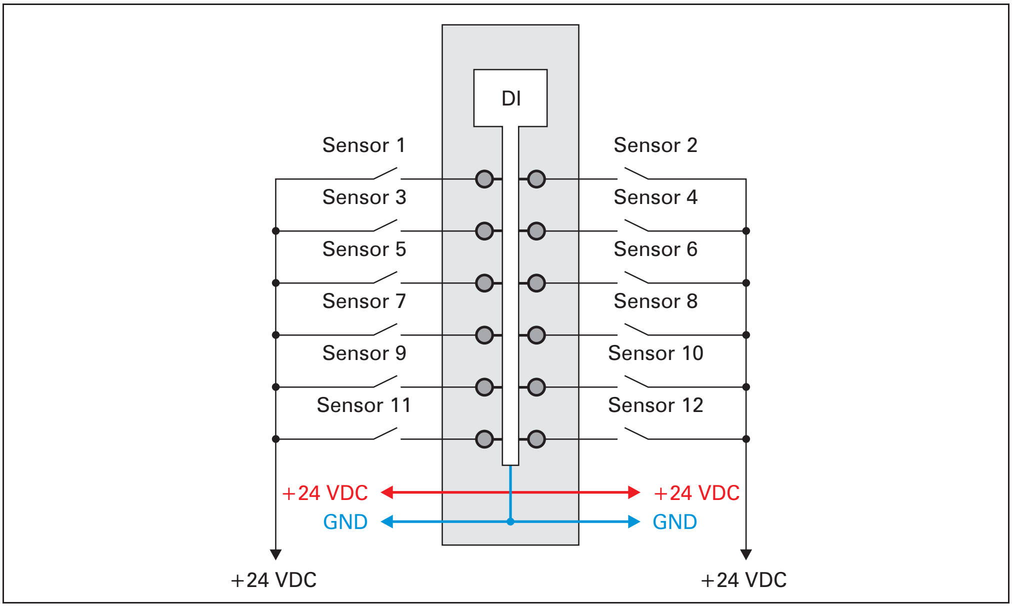


Figure 72: DI9371 - connection example

8.7.8 Input circuit diagram

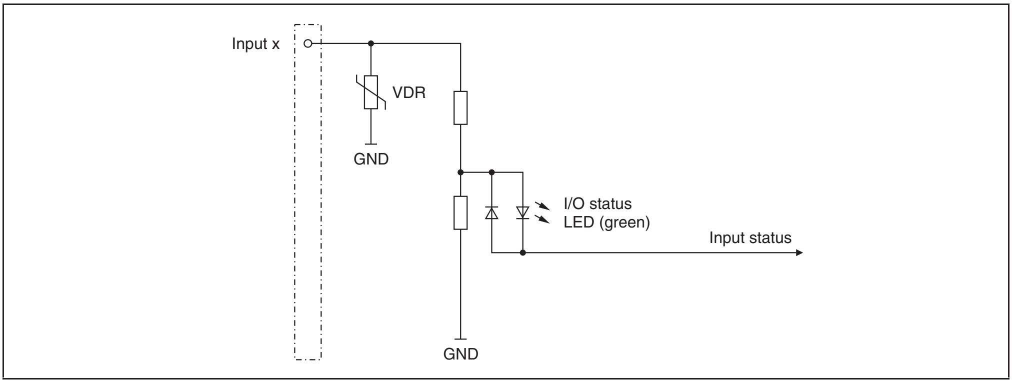


Figure 73: DI9371 - input circuit diagram

8.7.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 μs grid with a network-related jitter of up to 50 μs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.7.10 B&R ID code

Code for module identification ($1B95).

8.7.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering 1)	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.7.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

8.8 DI9372

8.8.1 General information

The DI9372 module is equipped with twelve inputs for 1-wire connections. The DI9372 designed for source input wiring.

- 12 digital inputs
- Source connection
- 1-wire connection
- Software input filter can be configured for entire module

8.8.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI9372	X20 digital input module, 12 inputs, 24 VDC, source, configurable input filter, 1-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 103: DI9372 - order data

8.8.3 Technical data

Product ID	DI9372
Short description	
I/O module	Twelve 24 VDC digital inputs for 1-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	 ≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Source
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	 Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.2 W Max. 1.56 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 104: DI9372 - technical data

8.8.4 Additional technical data

Product ID	DI9372
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	 <5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1D28

Table 105: DI9372 - additional technical data

8.8.5 Status LEDs

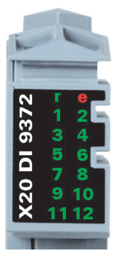
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 12	Green		Input status of the corresponding digital input

Table 106: DI9372 - status indicators

8.8.6 Pin assignments

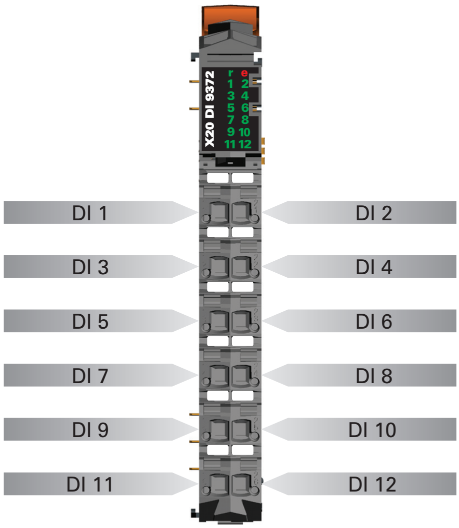


Figure 74: DI9372 - pin assignments

8.8.7 Connection example

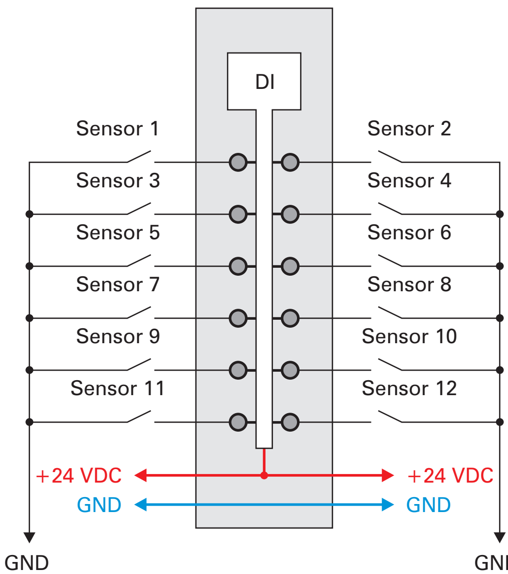


Figure 75: DI9372 - connection example

8.8.8 Input circuit diagram

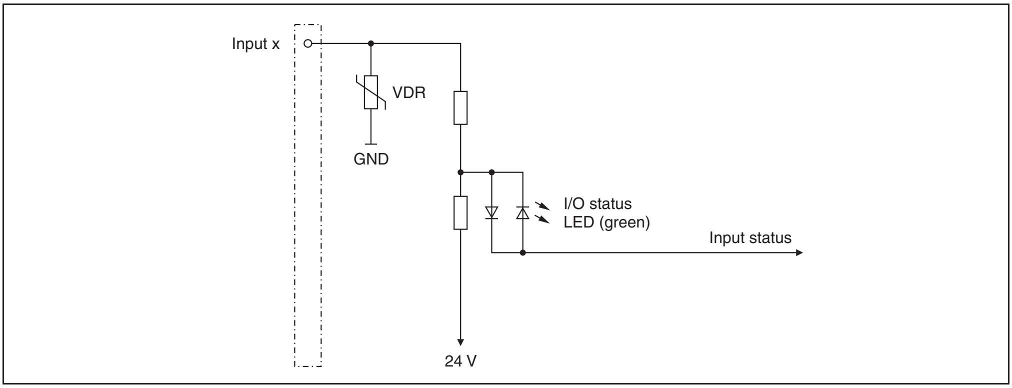


Figure 76: DI9372 - input circuit diagram

8.8.9 Digital inputs

Unfiltered

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

Filtered

The filtered status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle. Filtering takes place asynchronous to the network in a 200 µs grid with a network-related jitter of up to 50 µs.

The filter value can be configured for all digital inputs.

Value	Filter
0	No SW filter
2	0.2 ms
4	0.4 ms
:	:
250	25 ms - higher values are limited to this value

8.8.10 B&R ID code

Code for module identification ($1D28).

8.8.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Without filtering [1)]	≥100 µs

1) At cycle times of <150 µs, filtering is deactivated

8.8.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Without filtering	≥100 µs
With filtering	≥200 µs

9. Digital output modules

9.1 General information

Digital output modules are used to control external loads (relays, motors, solenoids). The states of the digital outputs are indicated by status LEDs.

9.1.1 Overview

Module	DO2322	DO4322	DO4332	DO6321	DO6322
Number of outputs	2	4	4	6	6
Rated voltage	24 VDC				
Rated output current	0.5 A	0.5 A	2.0 A	0.5 A	0.5 A
Total current	1.0 A	2.0 A	4.0 A	3.0 A	3.0 A
Output circuit	Source	Source	Source	Sink	Source
Connection type	3-wire connections	3-wire connections	3-wire connections	2-wire connections	2-wire connections

Table 107: Overview 1 of digital output modules

Module	DO8332	DO9321	DO9322
Number of outputs	8	12	12
Rated voltage	24 VDC		
Rated output current	2.0 A	0.5 A	0.5 A
Total current	8.0 A	6.0 A	6.0 A
Output circuit	Source	Sink	Source
Connection type	1-wire connections	1-wire connections	1-wire connections

Table 108: Overview 2 of digital output modules

9.2 DO2322

9.2.1 General information

The DO2322 module is equipped with two outputs for 3-wire connections. It is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 digital outputs
- Source connection
- 3-wire connection
- 24 VDC and GND for actuator supply
- Integrated output protection

9.2.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO2322	X20 digital output module, 2 outputs, 24 VDC, 0.5 A, source, 3-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 109: DO2322 - order data

9.2.3 Technical data

Product ID	DO2322
Short description	
I/O module	Two 24 VDC digital outputs for 3-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	1.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
Actuator supply	0.5 A in total for output-independent actuator supply
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.14 W Typ. 0.2 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 110: DO2322 - technical data

9.2.4 Additional technical data

Product ID	DO2322
Actuator supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Actuator supply	 Max. 12.0 W 1)
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.3 V @ 0.5 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.2.9 "Switching inductive loads" on page 208 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B96

Table 111: DO2322 - additional technical data

1) The power consumption of the sensors connected to the module may not exceed 12 W.

9.2.5 Status LEDs

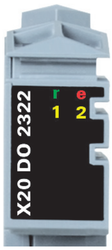
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Orange		Output status of the corresponding digital output

Table 112: DO2322 - status indicators

9.2.6 Pin assignments

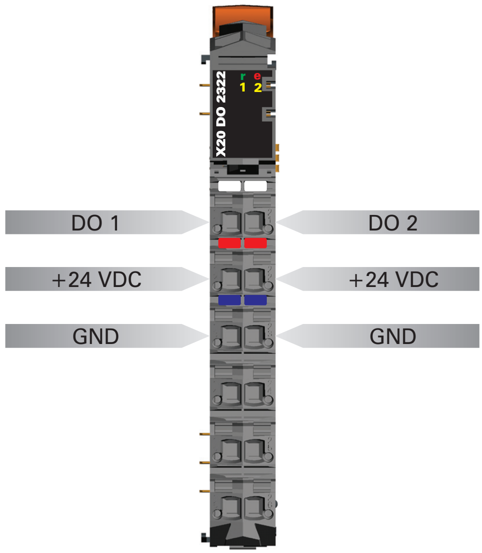


Figure 77: DO2322 - pin assignments

9.2.7 Connection example

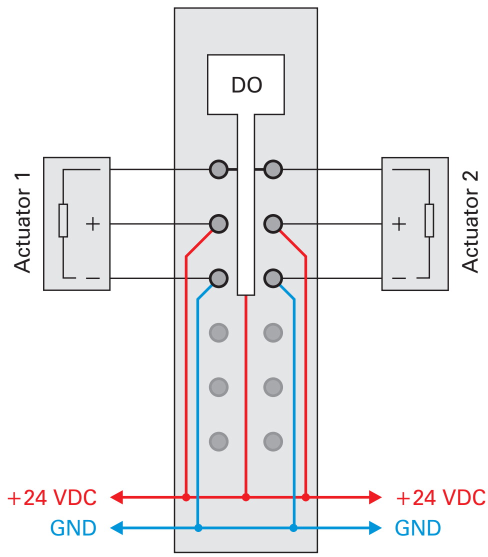


Figure 78: DO2322 - connection example

9.2.8 Output circuit diagram

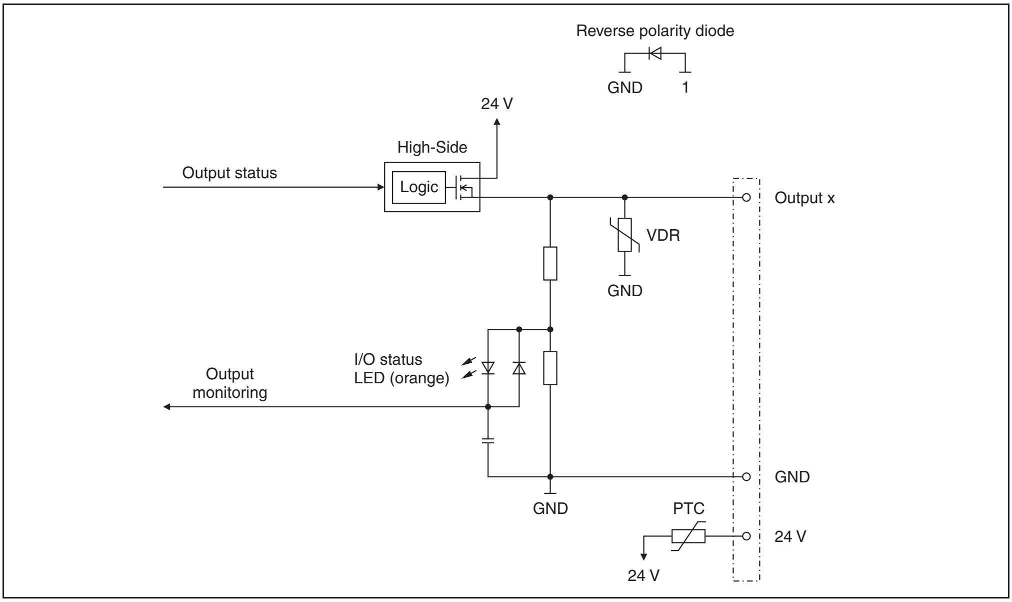


Figure 79: DO2322 - output circuit diagram

9.2.9 Switching inductive loads

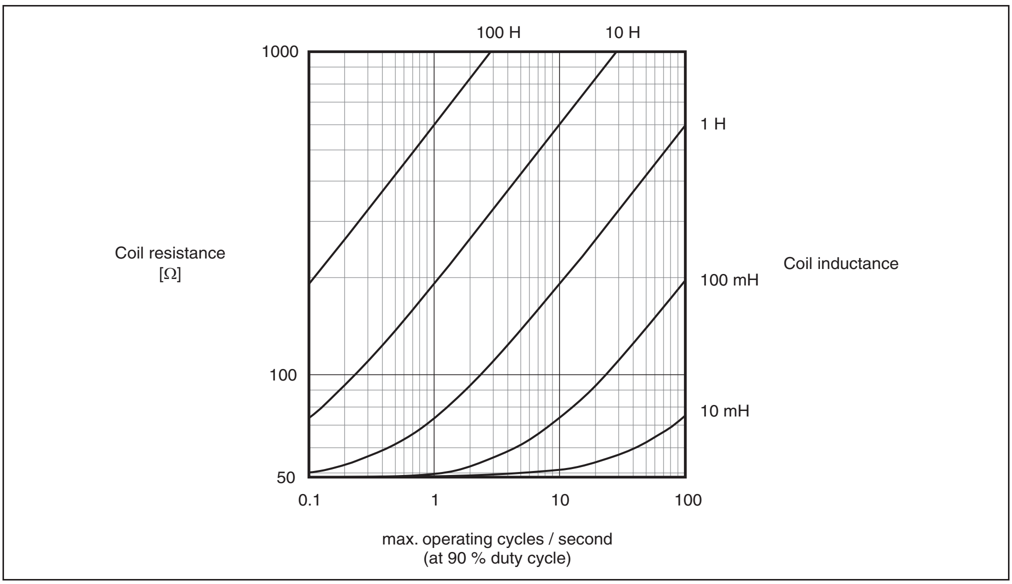


Figure 80: DO2322 - switching inductive loads

9.2.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.2.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.2.12 B&R ID code

Code for module identification ($1B96).

9.2.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.2.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.3 DO4322

9.3.1 General information

The DO4322 module is equipped with four outputs for 3-wire connections.

- 4 digital outputs
- Source connection
- 3-wire connection
- 24 VDC and GND for actuator supply
- Integrated output protection

9.3.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO4322	X20 digital output module, 4 outputs, 24 VDC, 0.5 A, source, 3-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 113: DO4322 - order data

9.3.3 Technical data

Product ID	DO4322
Short description	
I/O module	Four 24 VDC digital outputs for 3-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	2.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
Actuator supply	0.5 A in total for output-independent actuator supply
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.15 W Typ. 0.31 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 114: DO4322 - technical data

9.3.4 Additional technical data

Product ID	DO4322
Actuator supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Actuator supply	 Max. 12.0 W 1)
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.3 V @ 0.5 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.3.9 "Switching inductive loads" on page 215 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B97

Table 115: DO4322 - additional technical data

1) The power consumption of the sensors connected to the module may not exceed 12 W.

9.3.5 Status LEDs

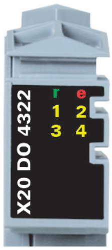
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Orange		Output status of the corresponding digital output

Table 116: DO4322 - status indicators

9.3.6 Pin assignments

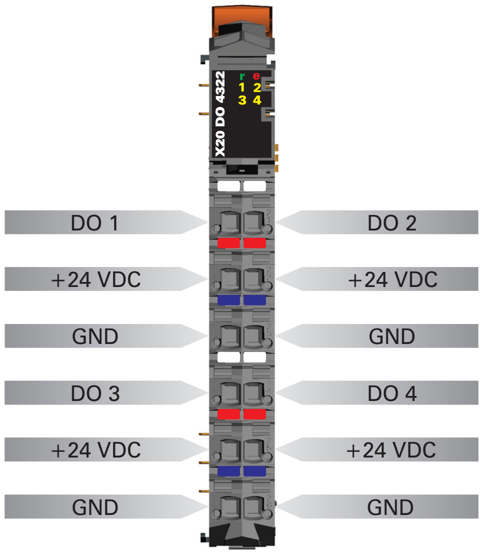


Figure 81: DO4322 - pin assignments

9.3.7 Connection example

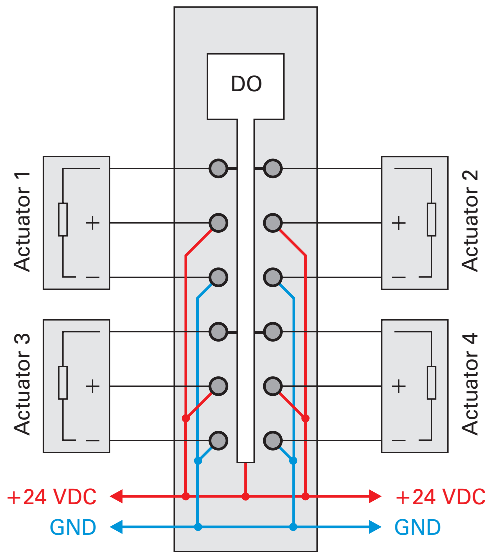


Figure 82: DO4322 - connection example

9.3.8 Output circuit diagram

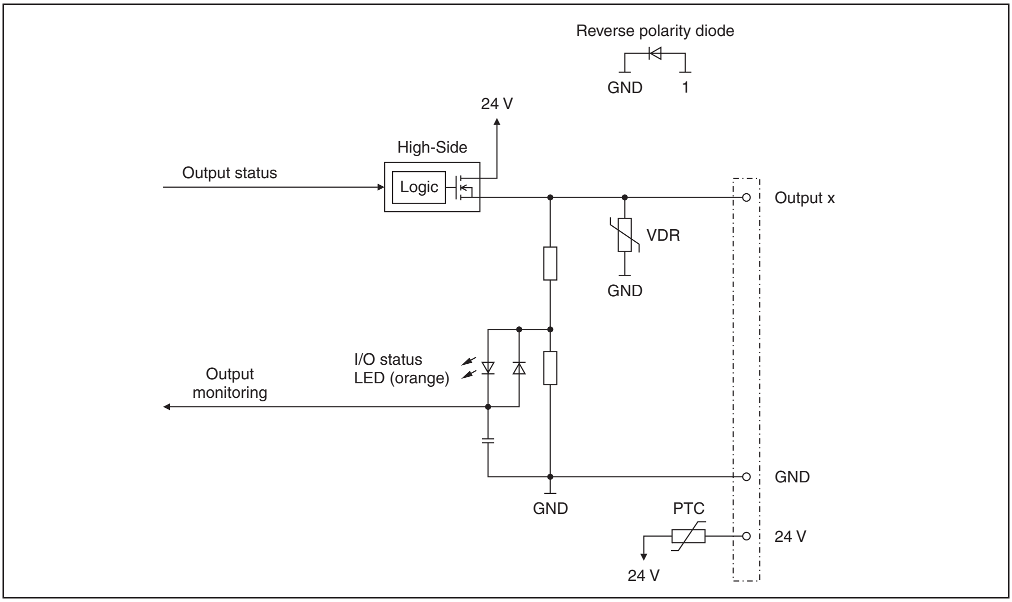


Figure 83: DO4322 - output circuit diagram

9.3.9 Switching inductive loads

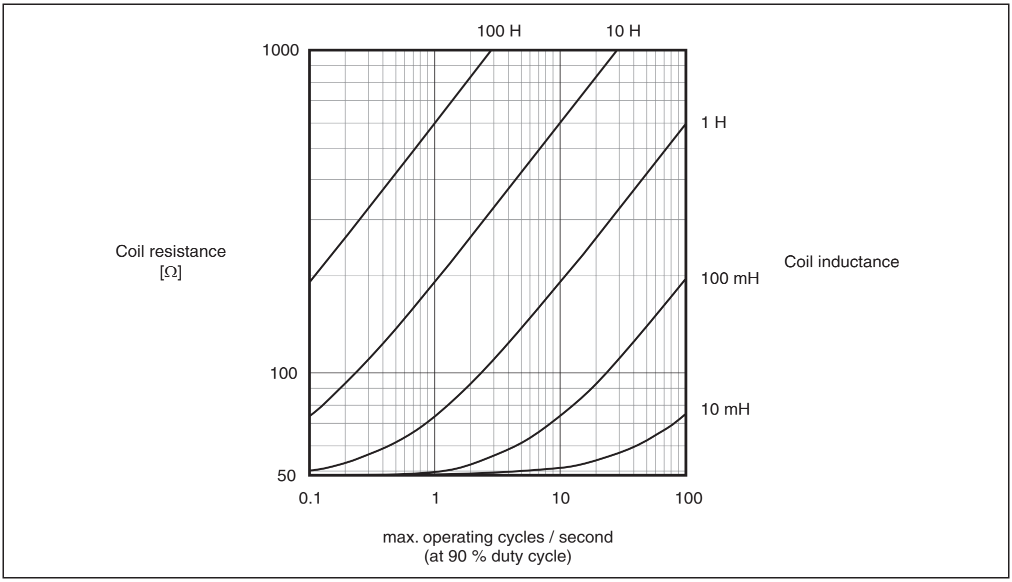


Figure 84: DO4322 - switching inductive loads

9.3.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.3.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.3.12 B&R ID code

Code for module identification ($1B97).

9.3.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.3.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.4 DO4332

9.4.1 General information

The DO4332 module is equipped with four outputs for 3-wire connections. The rated output current is 2.0 A.

- 4 digital outputs with 2 A
- Source connection
- 3-wire connection
- 24 VDC and GND for actuator supply
- Integrated output protection

9.4.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO4332	X20 digital output module, 4 outputs, 24 VDC, 2.0 A, source, 3-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 117: DO4332 - order data

9.4.3 Technical data

Product ID	DO4332
Short description	
I/O module	Four 24 VDC digital outputs for 3-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	2.0 A
Total current	4.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
Actuator supply	0.5 A in total for output-independent actuator supply
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.15 W Typ. 0.31 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 118: DO4332 - technical data

9.4.4 Additional technical data

Product ID	DO4332
Actuator supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Actuator supply	 Max. 12.0 W [1)]
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.5 V @ 2.0 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay $0 \rightarrow 1$ $1 \rightarrow 0$	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.4.9 "Switching inductive loads" on page 222 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B9C

Table 119: DO4332 - additional technical data

1) The power consumption of the sensors connected to the module may not exceed 12 W.

9.4.5 Status LEDs

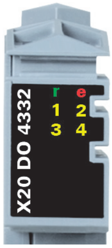
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Orange		Output status of the corresponding digital output

Table 120: DO4332 - status indicators

9.4.6 Pin assignments

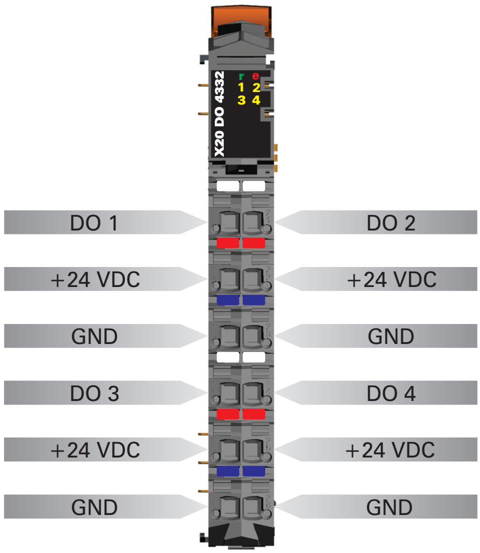


Figure 85: DO4332 - pin assignments

9.4.7 Connection example

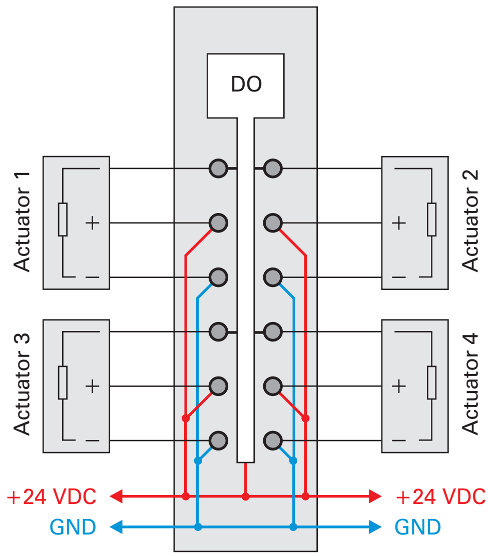


Figure 86: DO4332 - connection example

9.4.8 Output circuit diagram

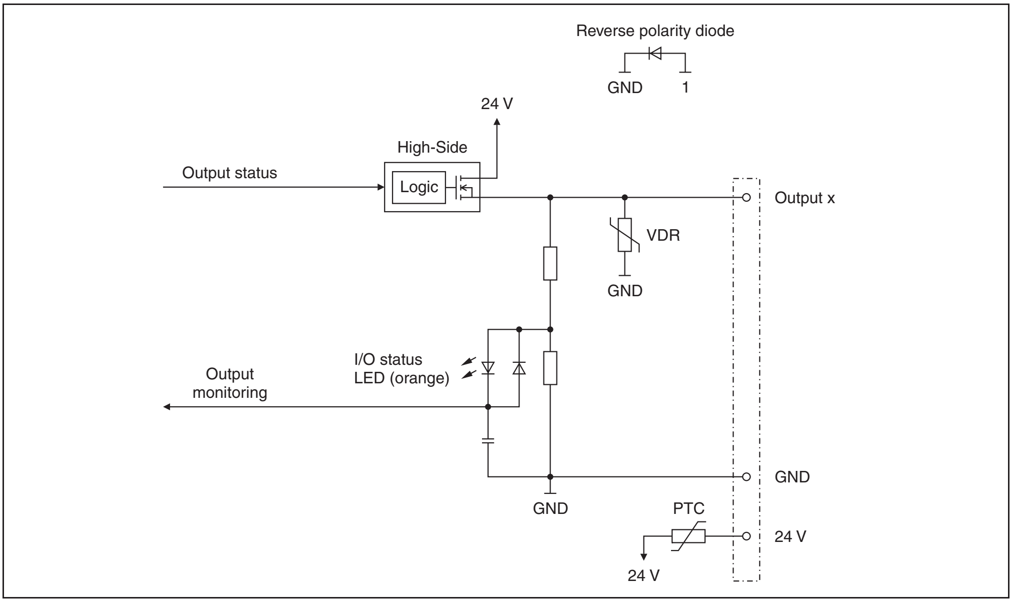


Figure 87: DO4332 - output circuit diagram

9.4.9 Switching inductive loads

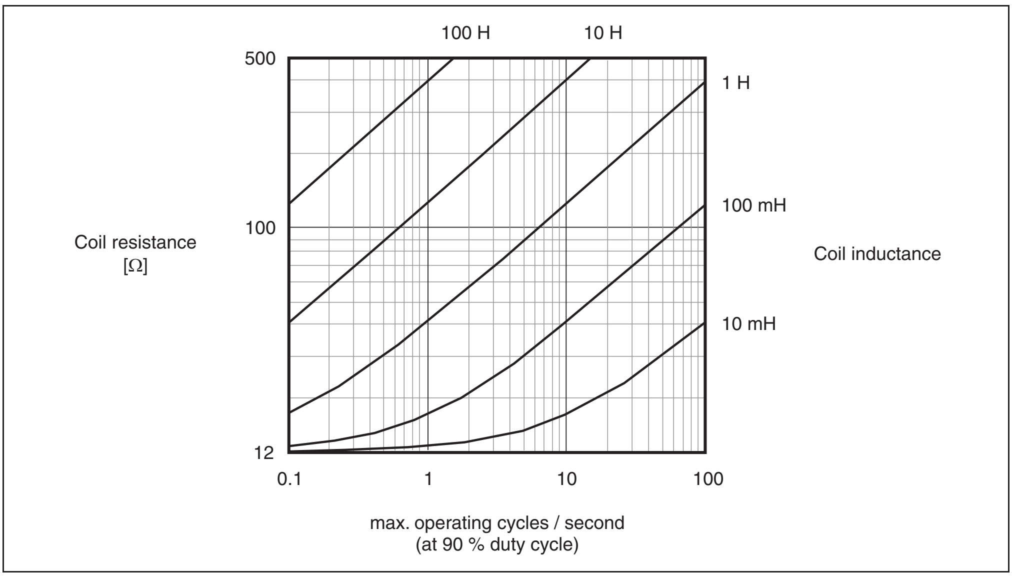


Figure 88: DO4332 - switching inductive loads

9.4.10 Operation with 2 A

The DO4332 outputs can handle up to 2 A. With a total current of 4 A, no more than 2 channels are operable at full load. Correct channel assignments are important for achieving optimal use of the module.

The following table provides an overview of the number of fully used channels and the resulting best distribution.

Number of channels using 2 A	Division
1	Any
2	The following channel numbers can be assigned: 1, 3 1, 4 2, 4

Table 121: DO4332 - operation with 2 A

9.4.11 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.4.12 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.4.13 B&R ID code

Code for module identification ($1B9C).

9.4.14 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 µs

9.4.15 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.5 DO6321

9.5.1 General information

The DO6321 module is equipped with six outputs for 1 or 2-wire connections. The X20 standard 6-pin terminal block can be used for universal 1-line wiring. Two-line wiring can be implemented using the 12-pin terminal block. The DO6321 is designed for sink output wiring.

- 6 digital outputs
- Sink connection
- 2-wire connection
- 24 VDC for signal supply
- Integrated output protection
- 1-wire connection type with 6-pin terminal block

9.5.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO6321	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, sink, 2-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 122: DO6321 - order data

9.5.3 Technical data

Product ID	DO6321
Short description	
I/O module	Six 24 VDC digital outputs for 1 or 2-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	3.0 A
Output circuit	Sink
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.16 W Typ. 0.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 123: DO6321 - technical data

9.5.4 Additional technical data

Product ID	DO6321
Digital outputs	
Type	FET negative switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	75 µA
Short circuit peak current	<7 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.5.9 "Switching inductive loads" on page 229 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B99

Table 124: DO6321 - additional technical data

9.5.5 Status LEDs

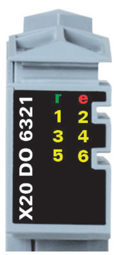
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 6	Orange		Output status of the corresponding digital output

Table 125: DO6321 - status indicators

9.5.6 Pin assignments

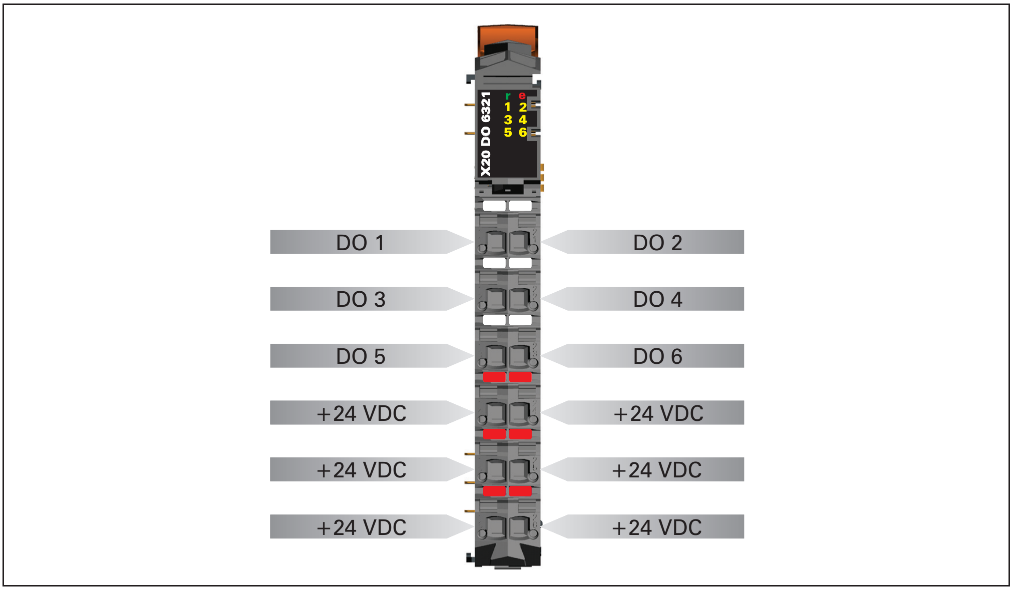


Figure 89: DO6321 - pin assignments

9.5.7 Connection example

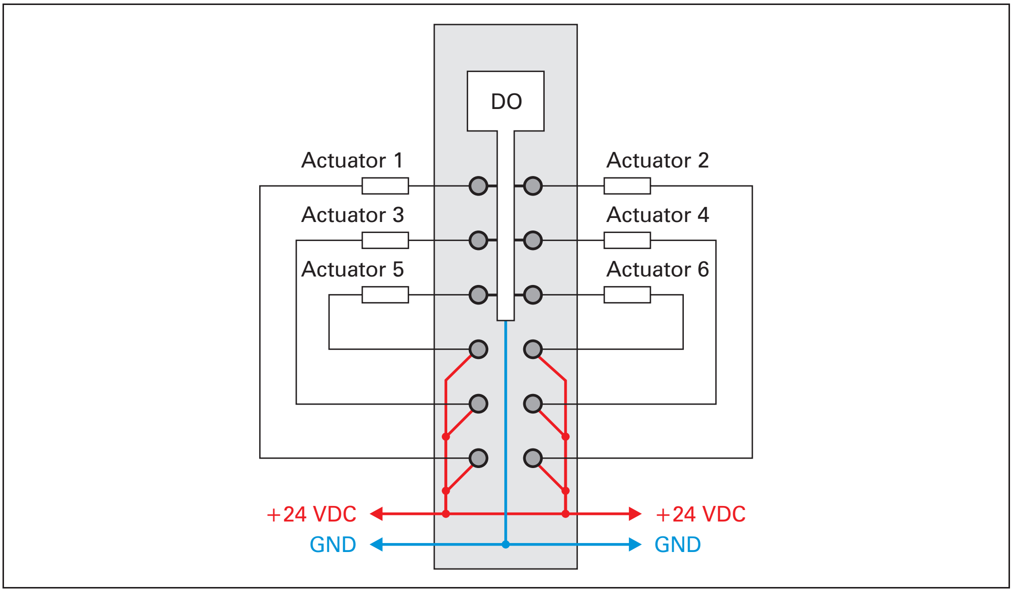


Figure 90: DO6321 - connection example

9.5.8 Output circuit diagram

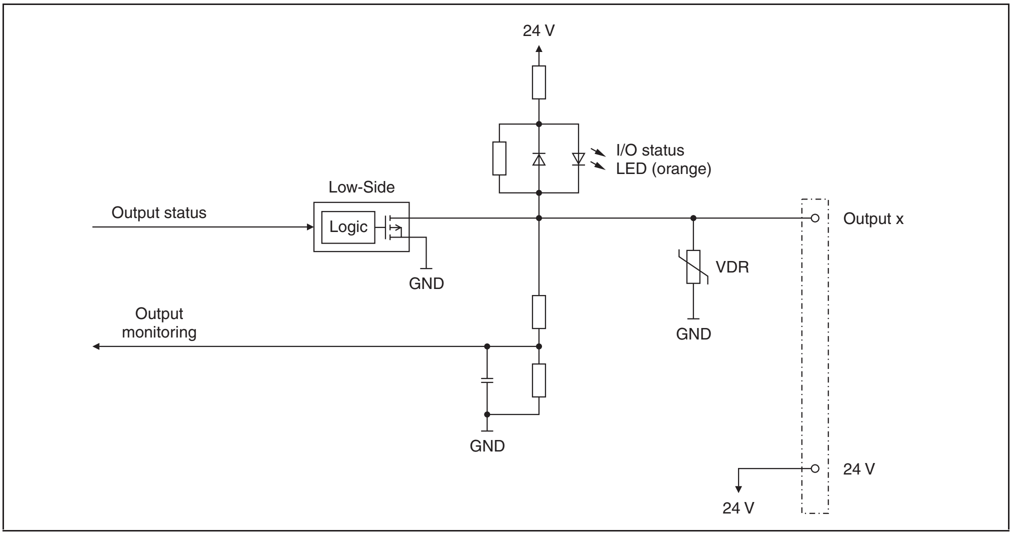


Figure 91: DO6321 - output circuit diagram

9.5.9 Switching inductive loads

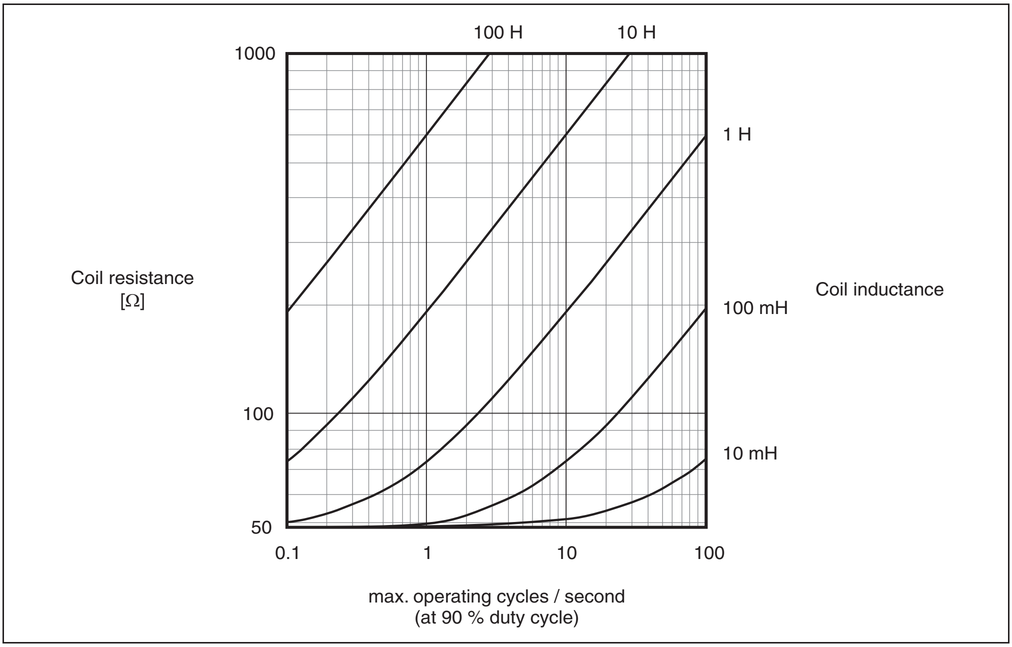


Figure 92: DO6321 - switching inductive loads

9.5.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.5.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.5.12 B&R ID code

Code for module identification ($1B99).

9.5.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.5.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.6 DO6322

9.6.1 General information

The DO6322 module is equipped with six outputs for 1 or 2-wire connections. The X20 standard 6-pin terminal block can be used for universal 1-line wiring. Two-line wiring can be implemented using the 12-pin terminal block. The DO6322 designed for source output wiring.

- 6 digital outputs
- Source connection
- 2-wire connection
- GND for signal supply
- Integrated output protection
- 1-wire connection type with 6-pin terminal block

9.6.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO6322	X20 digital output module, 6 outputs, 24 VDC, 0.5 A, source, 2-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 126: DO6322 - order data

9.6.3 Technical data

Product ID	DO6322
Short description	
I/O module	Six 24 VDC digital outputs for 1 or 2-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	3.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.16 W Typ. 0.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 127: DO6322 - technical data

9.6.4 Additional technical data

Product ID	DO6322
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.3 V @ 0.5 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.6.9 "Switching inductive loads" on page 235 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B98

Table 128: DO6322 - additional technical data

9.6.5 Status LEDs

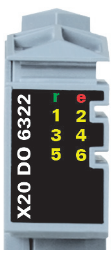
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 6	Orange		Output status of the corresponding digital output

Table 129: DO6322 - status indicators

9.6.6 Pin assignments

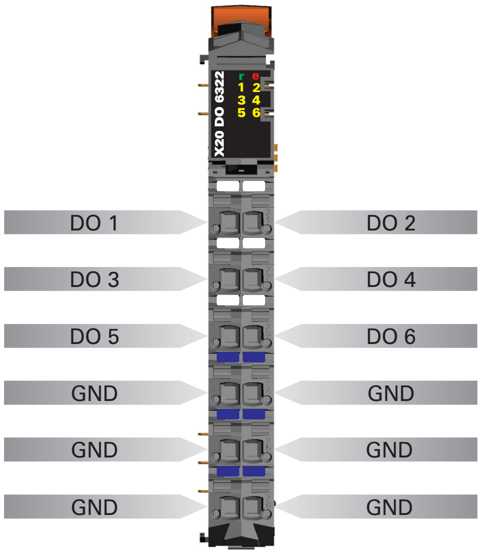


Figure 93: DO6322 - pin assignments

9.6.7 Connection example

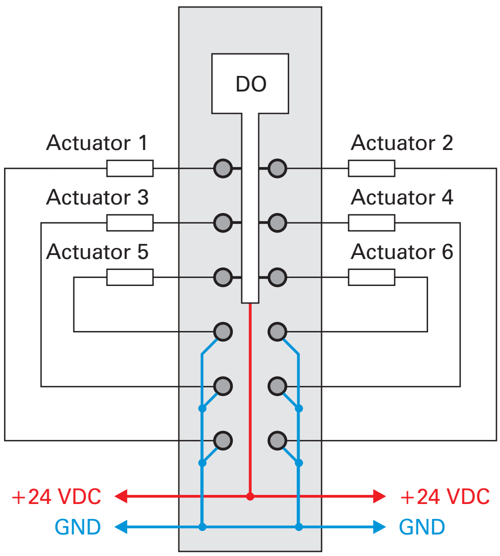


Figure 94: DO6322 - connection example

9.6.8 Output circuit diagram

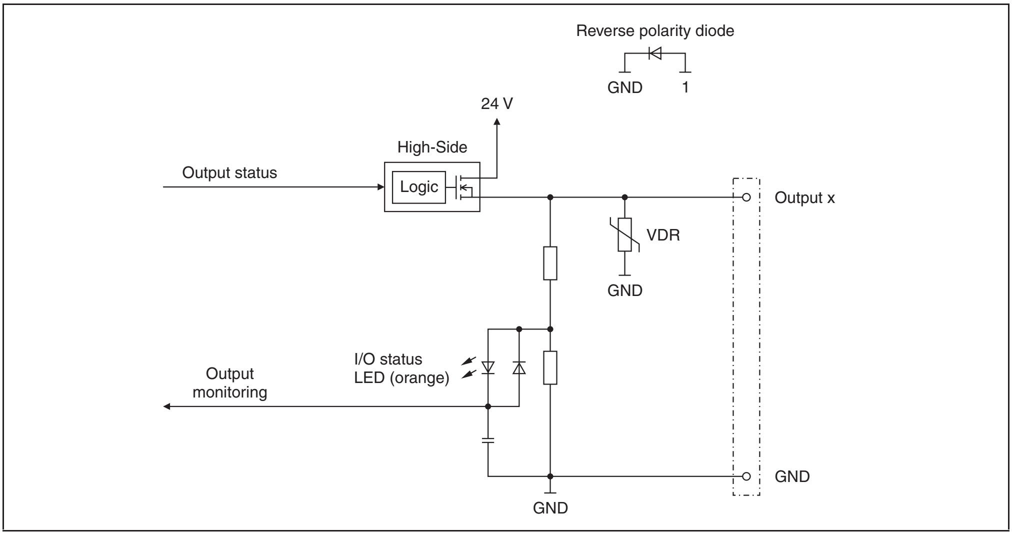


Figure 95: DO6322 - output circuit diagram

9.6.9 Switching inductive loads

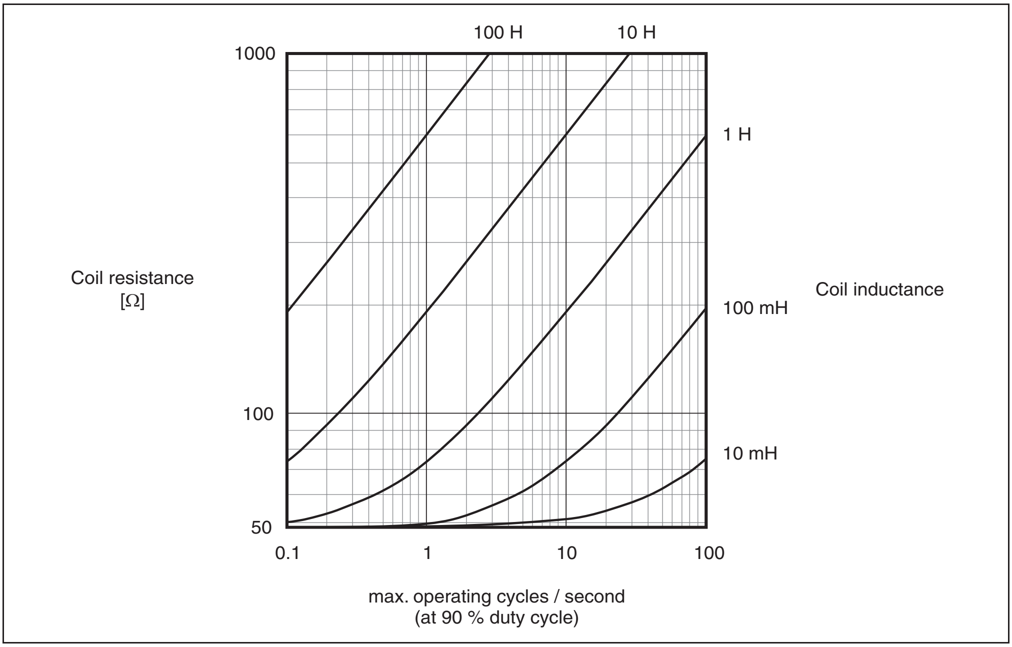


Figure 96: DO6322 - switching inductive loads

9.6.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.6.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.6.12 B&R ID code

Code for module identification ($1B98).

9.6.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.6.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.7 DO8332

9.7.1 General information

The DO8332 module is equipped with eight outputs for 1-wire connections. The rated output current is 2 A.

The output supply is fed directly on the module. An additional supply module is not needed. There is no connection between the module and the I/O supply potential on the bus module.

- 8 digital outputs with 2 A
- Source connection
- 1-wire connection
- Power feed integrated in the module
- Integrated output protection

9.7.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO8332	X20 digital output module, 8 outputs, 24 VDC, 2.0 A, source, feed directly on module, 1-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 130: DO8332 - order data

9.7.3 Technical data

Product ID	DO8332
Short description	
I/O module	Eight 24 VDC digital outputs for 1-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	2.0 A
Total current	8.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances, reverse polarity protection of the supply protection.
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal I/O external	 Typ. 0.21 W - Typ. 0.69 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 131: DO8332 - technical data

9.7.4 Additional technical data

Product ID	DO8332
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.5 V @ 2.0 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.7.9 "Switching inductive loads" on page 241 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B9D

Table 132: DO8332 - additional technical data

9.7.5 Status LEDs

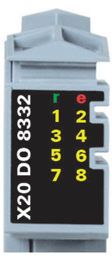
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
			Double flash	I/O supply too low
	e + r	Steady red / single green flash		Invalid firmware
	1 - 8	Orange		Output status of the corresponding digital output

Table 133: DO8332 - status indicators

9.7.6 Pin assignments

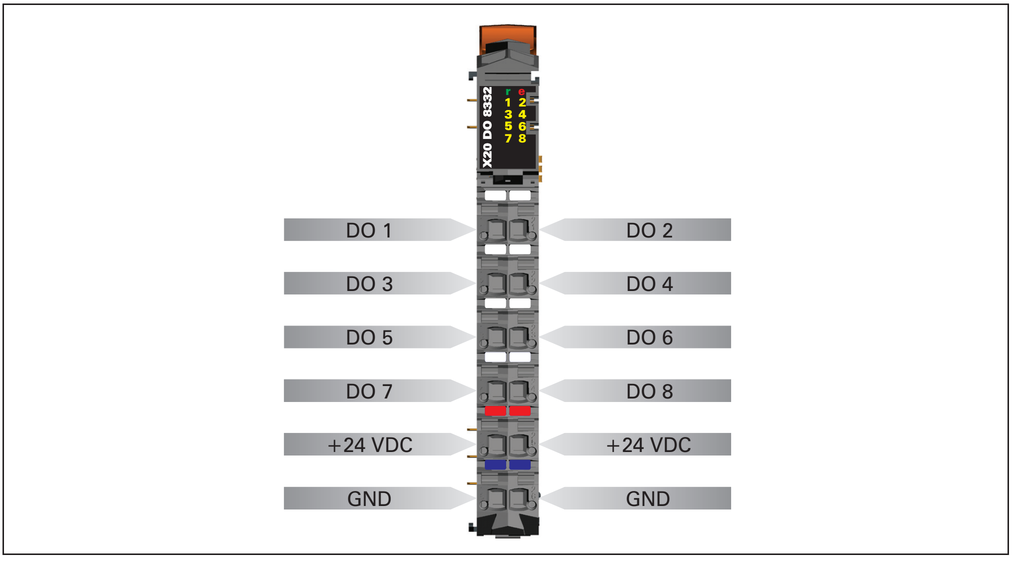


Figure 97: DO8332 - pin assignments

9.7.7 Connection example

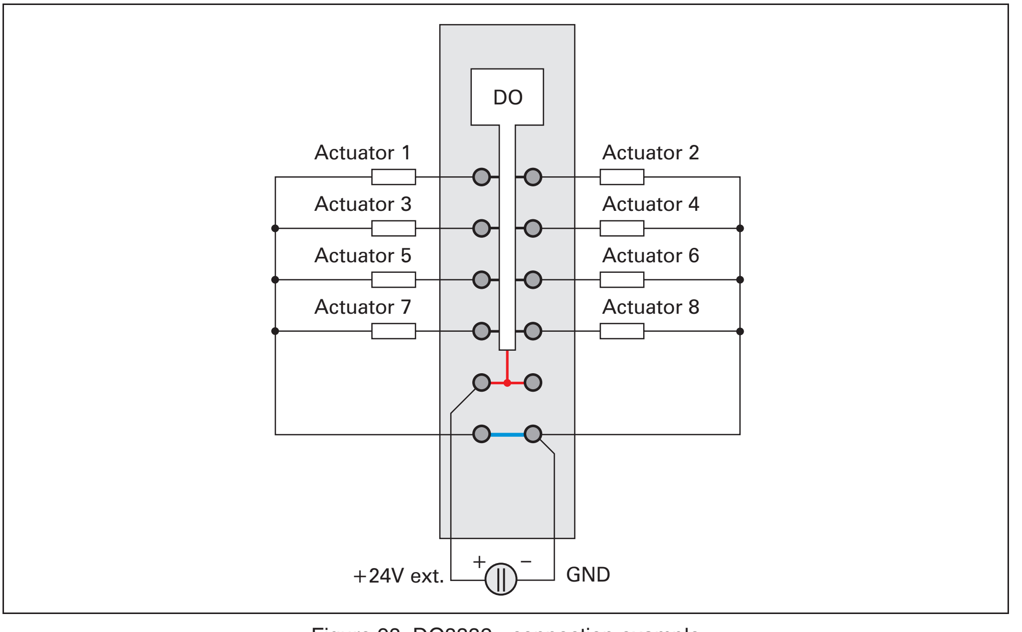


Figure 98: DO8332 - connection example

9.7.8 Output circuit diagram

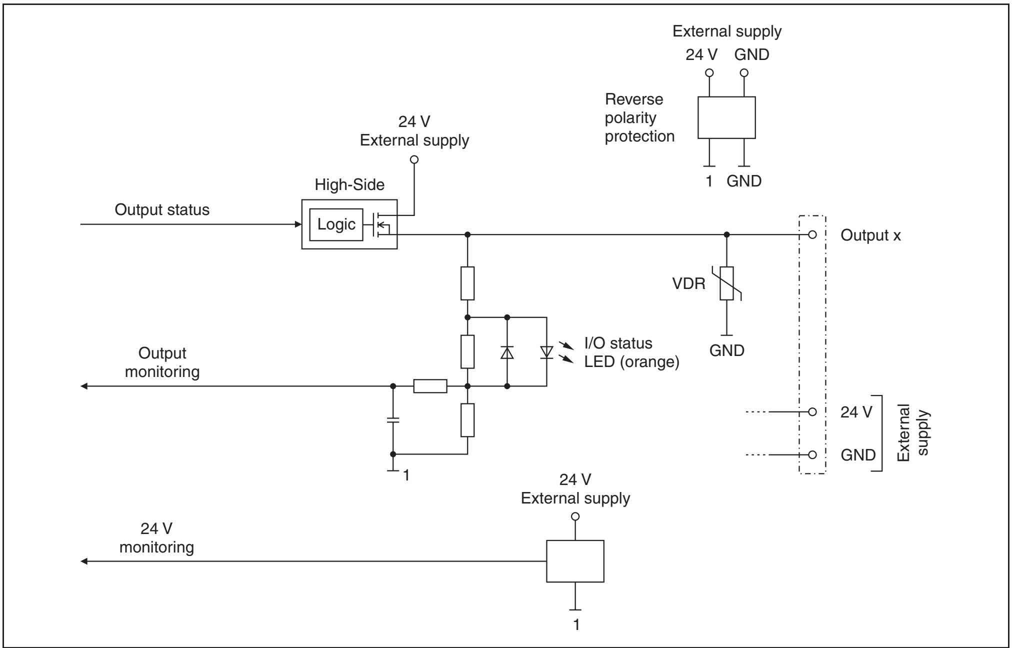


Figure 99: DO8332 - output circuit diagram

9.7.9 Switching inductive loads

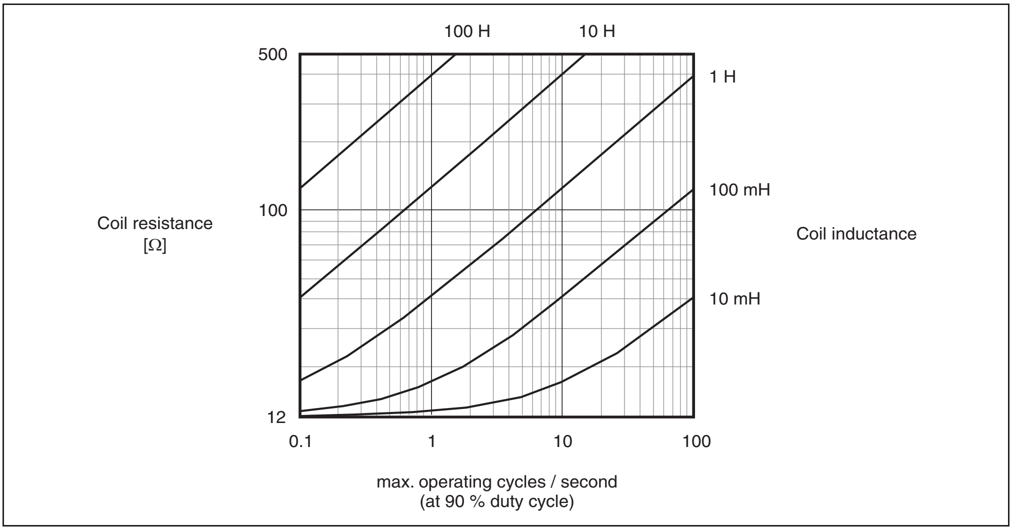


Figure 100: DO8332 - switching inductive loads

9.7.10 Operation with 2 A

The DO8332 outputs can handle up to 2 A. With a total current of 8 A, no more than 4 channels are operable at full load. To ensure optimal use of the module, it is important to assign the channels properly, and to keep in mind a potential derating.

Correct channel assignment is important, since the eight outputs are divided between two output drivers. The channels operated with 2 A must therefore be evenly divided between both output drivers.

Output driver 1: channels 1 - 4
Output driver 2: channels 5 - 8

The following table provides an overview of the number of fully used channels, the resulting best distribution, and a potential derating.

Number of channels using 2 A	Division	Derating
1	Any	No
2	1. Channel with 2 A ... Channel no. 1 - 4 2. Channel with 2 A ... Channel no. 5 - 8	No
3	Assign all even or all odd channel numbers. Examples: 1, 3, 5 2, 4, 6 3, 5, 7 4, 6, 8	 Channels 1 and 3 Channels 2 and 4 Channels 5 and 7 Channels 6 and 8
4	Assign all even or all odd channel numbers. Possible distributions: 1, 3, 5, 7 2, 4, 6, 8	 All channels All channels

Table 134: DO8332 - operation with 2 A

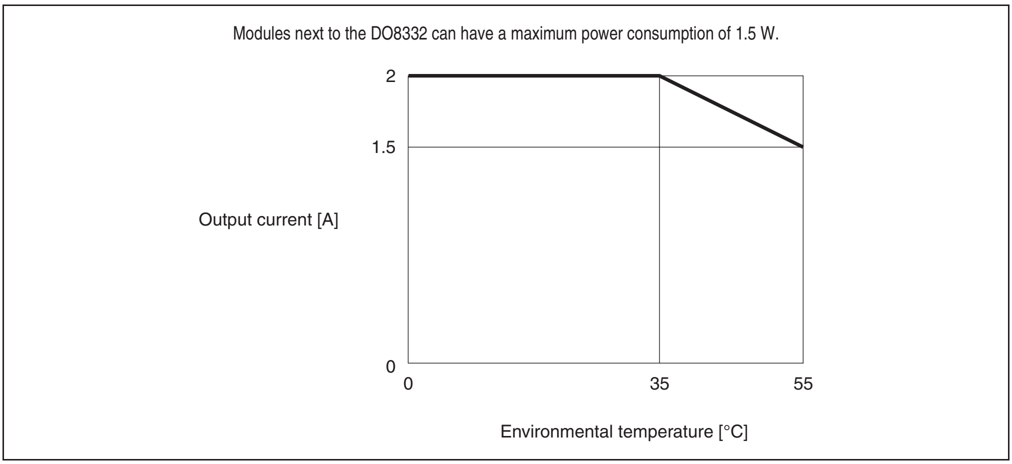


Figure 101: DO8332 - derating when 3 or 4 channels are operated with 2 A

9.7.11 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.7.12 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.7.13 Operating limit monitoring

The module's output supply is monitored. An I/O supply voltage of <20.4 V is displayed as a warning.

9.7.14 Additional function - switch digital outputs with delay using switching mask

In function model 1 the digital outputs can be operated with a timer on a 100 µs basis and with your choice of switching mask. After the delay time has expired, the digital outputs are adjusted according to the switching mask and the delayed output pattern.

Sequence

1) The delay time is changed or set to 0:

```
Output status = status is taken from register 2
```

2) After delay time has expired:

```
Output status = ( (register 2 AND INV(register 6)) OR
                (register 4 AND register 6) )

Register 2 = "DigitalOutput 1 - 8" register or
             "DigitalOutput01" register - "DigitalOutput08" register

Register 4 = "DigitalOutput delayed 1 - 8" register or
             "DigitalOutput01Delayed"  register  -  "DigitalOutput08Delayed"
register

Register 6 = "DigitalOutput switching mask 1 - 8" register or
             register
             "DigitalOutput01DelayEnable" register -
"DigitalOutput08DelayEnable" register
```

9.7.15 B&R ID code

Code for module identification ($1B9D).

9.7.16 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Function model 0	≥100 μs
Function model 1	≥150 μs

9.7.17 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Function model 0	Equal to the minimum cycle time
Function model 1	Equal to the minimum cycle time

9.8 DO9321

9.8.1 General information

The DI9371 module is equipped with twelve outputs for 1-wire connections. The DO9321 is designed for sink output wiring.

- 12 digital outputs
- Sink connection
- 1-wire connection
- Integrated output protection

9.8.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO9321	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, sink, 1-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 135: DO9321 - order data

9.8.3 Technical data

Product ID	DO9321
Short description	
I/O module	Twelve 24 VDC digital outputs for 1-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	6.0 A
Output circuit	Sink
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.21 W Typ. 0.99 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 136: DO9321 - technical data

9.8.4 Additional technical data

Product ID	DO9321
Digital outputs	
Type	FET negative switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	75 µA
Short circuit peak current	<7 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.8.9 "Switching inductive loads" on page 250 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B9B

Table 137: DO9321 - additional technical data

9.8.5 Status LEDs

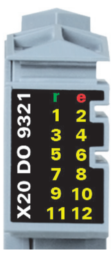
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 12	Orange		Output status of the corresponding digital output

Table 138: DO9321 - status indicators

9.8.6 Pin assignments

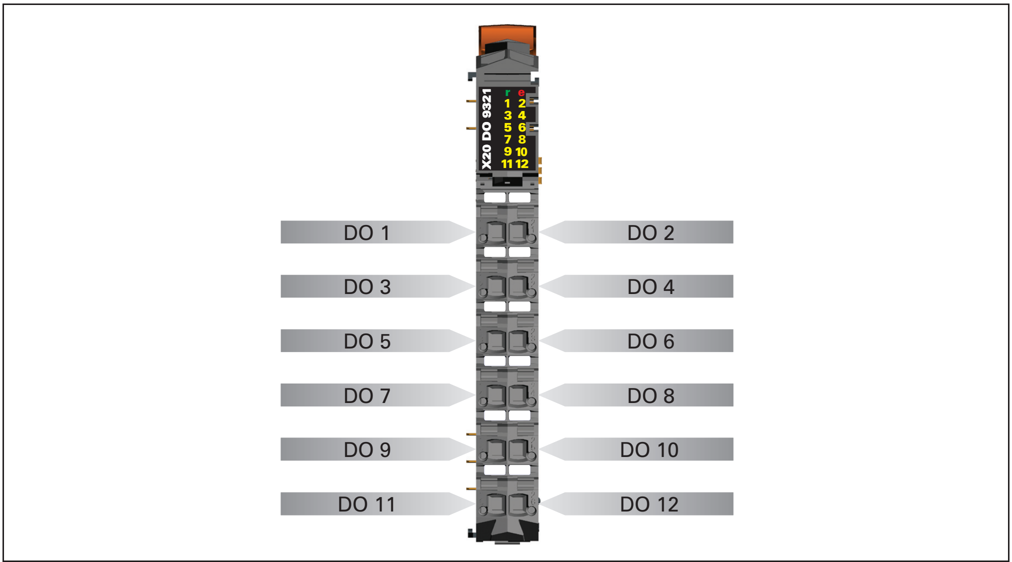


Figure 102: DO9321 - pin assignment

9.8.7 Connection example

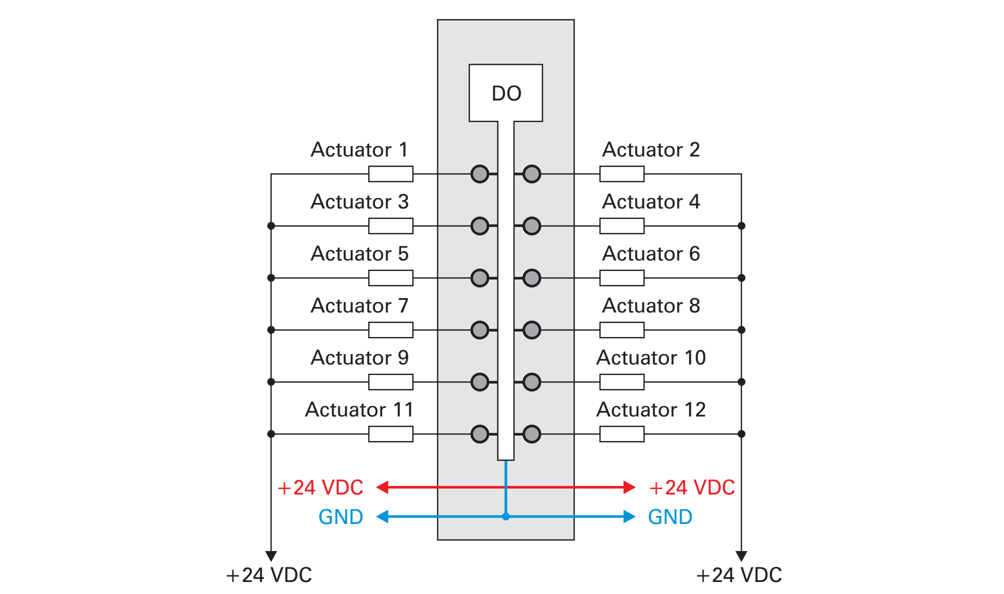


Figure 103: DO9321 - connection example

9.8.8 Output circuit diagram

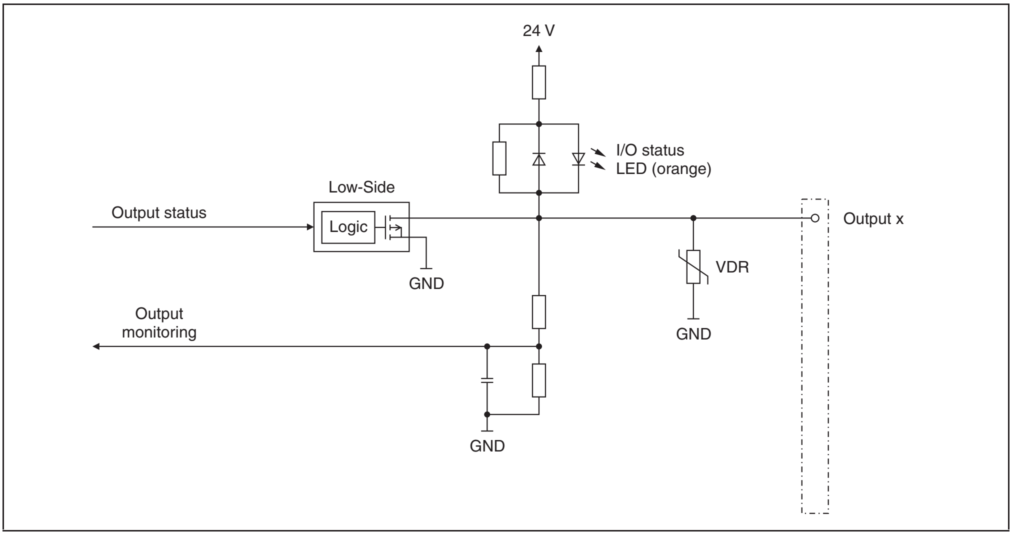


Figure 104: DO9321 - output circuit diagram

9.8.9 Switching inductive loads

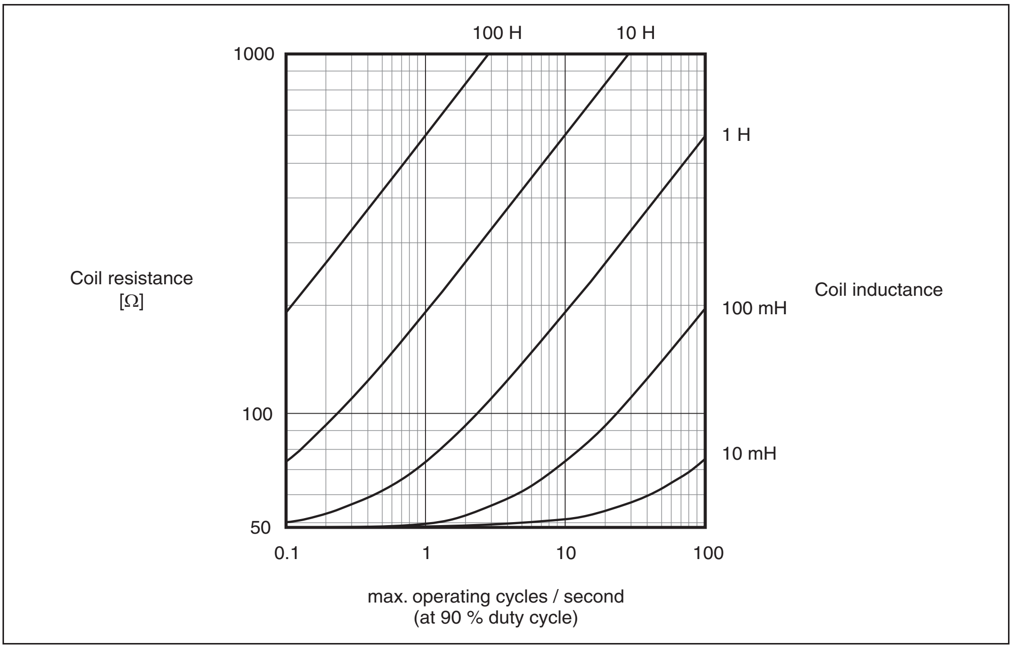


Figure 105: DO9321 - switching inductive loads

9.8.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.8.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.8.12 B&R ID code

Code for module identification ($1B9B).

9.8.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.8.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

9.9 DO9322

9.9.1 General information

The DO9322 module is equipped with twelve outputs for 1-wire connections. The DO9322 designed for source output wiring.

- 12 digital outputs
- Source connection
- 1-wire connection
- Integrated output protection

9.9.2 Order data

Model number	Short description	Image
	Digital output module	
X20DO9322	X20 digital output module, 12 outputs, 24 VDC, 0.5 A, source, 1-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 139: DO9322 - order data

9.9.3 Technical data

Product ID	DO9322
Short description	
I/O module	Twelve 24 VDC digital outputs for 1-wire connections
Digital outputs	
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	6.0 A
Output circuit	Source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	 Yes, with status LED and software status Yes, with status LED and software status (output error status)
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.21 W Typ. 0.99 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 140: DO9322 - technical data

9.9.4 Additional technical data

Product ID	DO9322
Digital outputs	
Type	FET positive switching
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring with 10 ms delay
Leakage current when switched off	5 µA
Residual voltage	<0.3 V @ 0.5 A rated current
Short circuit peak current	<12 A
Switching on after overload or short circuit cutoff	Approx. 10 ms (depends on the module temperature)
Switching delay $0 \rightarrow 1$ $1 \rightarrow 0$	 <300 µs <300 µs
Switching frequency Resistive load Inductive load	 Max. 500 Hz See section 9.9.9 "Switching inductive loads" on page 256 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Typ. 50 VDC
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1B9A

Table 141: DO9322 - additional technical data

9.9.5 Status LEDs

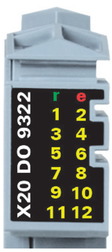
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Single flash	Warning / error for an I/O channel. Level monitoring for digital outputs has responded.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 12	Orange		Output status of the corresponding digital output

Table 142: DO9322 - status indicators

9.9.6 Pin assignments

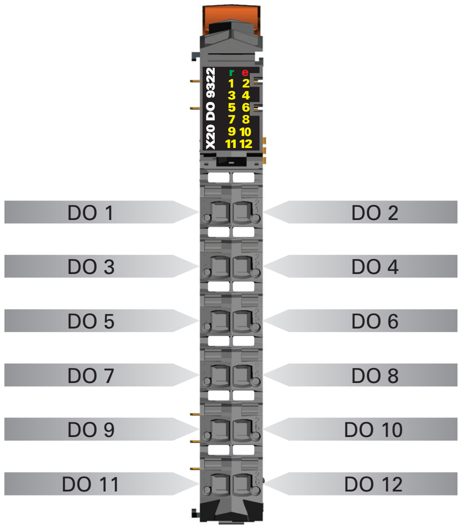


Figure 106: DO9322 - pin assignments

9.9.7 Connection example

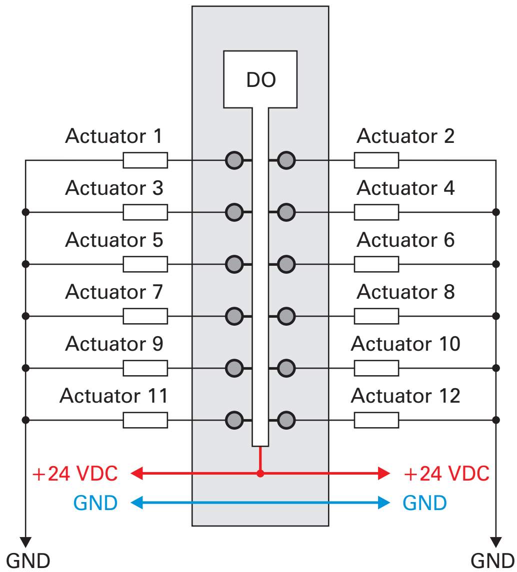


Figure 107: DO9322 - connection example

9.9.8 Output circuit diagram

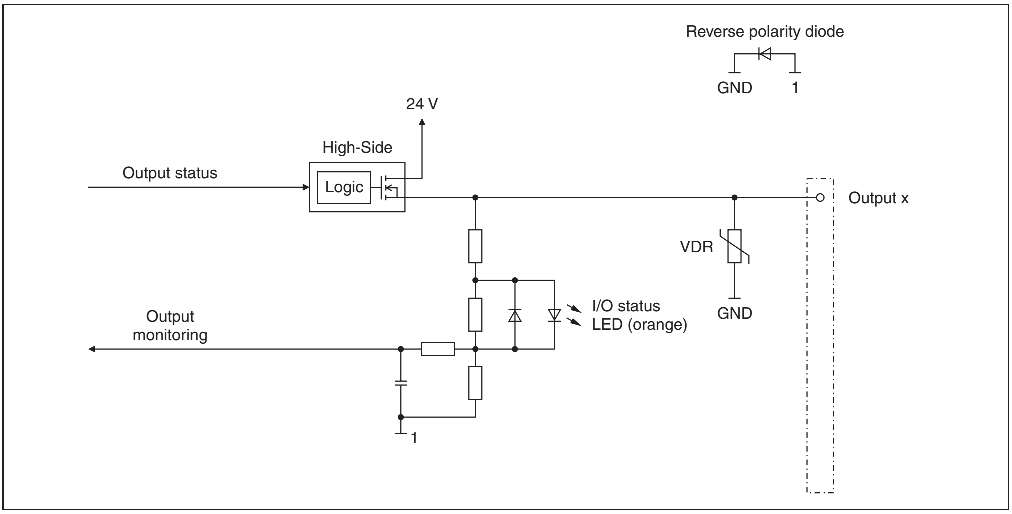


Figure 108: DO9322 - output circuit diagram

9.9.9 Switching inductive loads

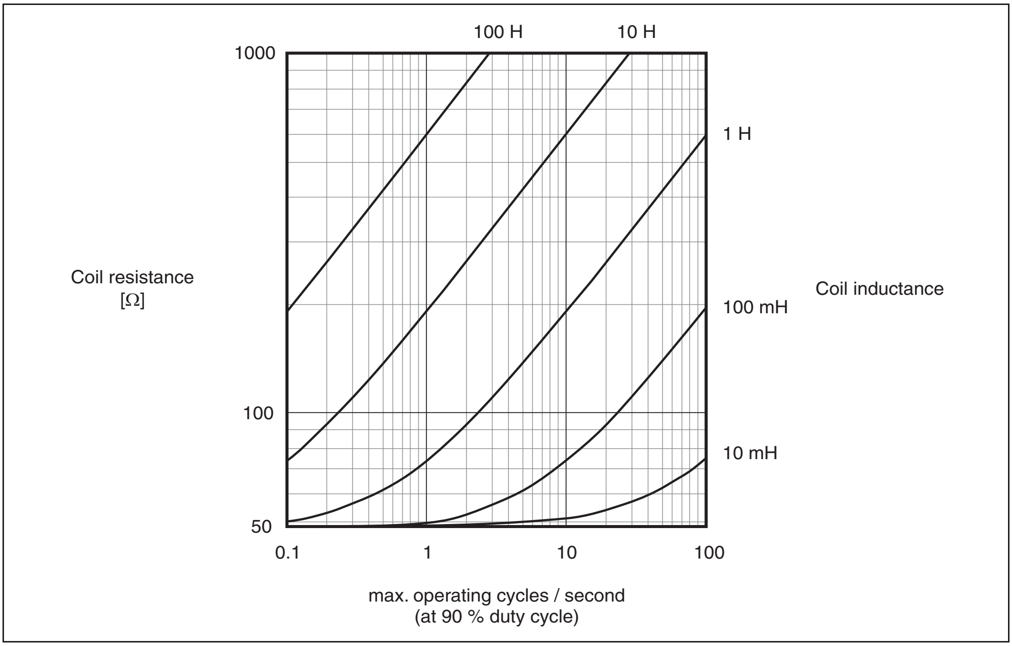


Figure 109: DO9322 - switching inductive loads

9.9.10 Digital outputs

The output status is transferred to the output channels using a fixed offset to the network cycle.

9.9.11 Output monitoring status

The output states for the outputs are compared to the set values on the module. The control for the output driver is used for the set states.

A change in the output status resets the monitoring for this output. The status of each individual channel can be read. A change in the monitoring status generates an error message.

Monitoring status	Description
0	Digital output channel: No error
1	Digital output channel: Short circuit or overload

9.9.12 B&R ID code

Code for module identification ($1B9A).

9.9.13 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

9.9.14 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
Equal to the minimum cycle time

10. Analog input modules

10.1 General information

Analog input modules convert measured values (voltages, currents) into numerical values, which can be processed by the PLC.

In the PLC, analog data is always in 16-bit 2s complement regardless of the resolution. Therefore, the resolution of the module used does not have to be taken into consideration when creating an application program.

Every channel on an analog input module has a status LED.

10.1.1 Overview

Module	AI2622	AI2632	AI4622	AI4632
Number of inputs	2	2	4	4
Input signal, rated	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]
Digital converter resolution	±12-bit	16-bit	±12-bit	16-bit

Table 143: Overview of analog input modules

1) Using different connection terminals.

10.1.2 Input filter

All modules are equipped with a configurable input filter. The minimum cycle time must be >500 μs. Filtering is deactivated for shorter cycle times.

If the input filter is active, then the channels are scanned in ms cycles. The time offset between the channels is 200 μs. The conversion takes place asynchronously to the network cycle.

Input ramp limitation

Input ramp limitation can only take place together with filtering. And input ramp limitation is executed before filtering takes place.

The difference of the change in the input value is checked to make sure the specified limits are not exceeded. If the values are exceeded, the adjusted input value is equal to the old value ± the limit value.

Adjustable limit values:

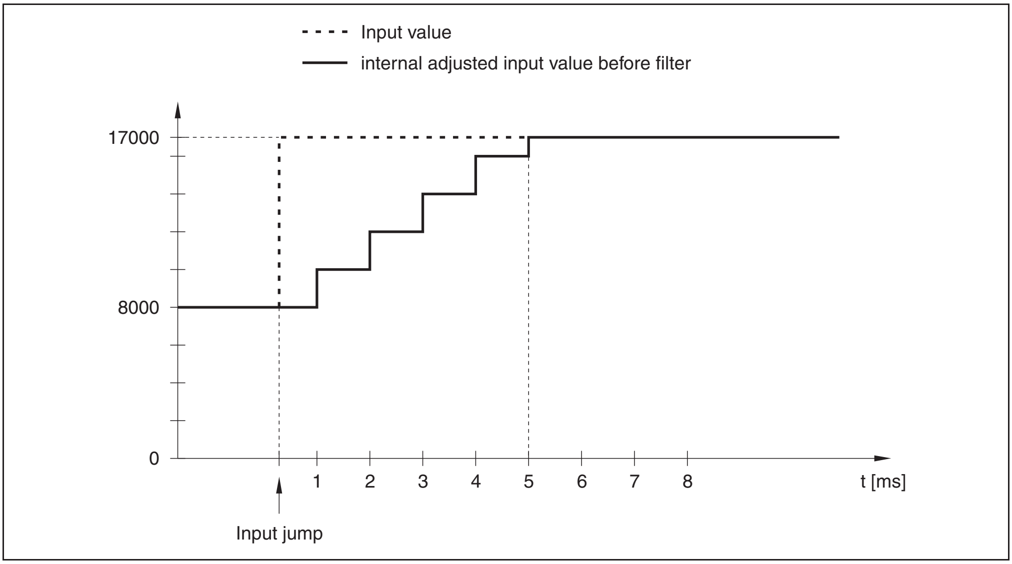
Code	Limit value
0	The input value is used without limitation.
1	$3FFF = 16383
2	$1FFF = 8191
3	$0FFF = 4095
4	$07FF = 2047
5	$03FF = 1023
6	$01FF = 511
7	$00FF = 255

The input ramp limitation is well suited for suppressing disturbances (spikes). The following examples show the function of the input ramp limitation based on an input jump and a disturbance.

Example 1: The input value makes a jump from 8,000 to 17,000. The diagram displays the adjusted input value for the following settings:

Input ramp limitation = 4 = $07FF = 2047

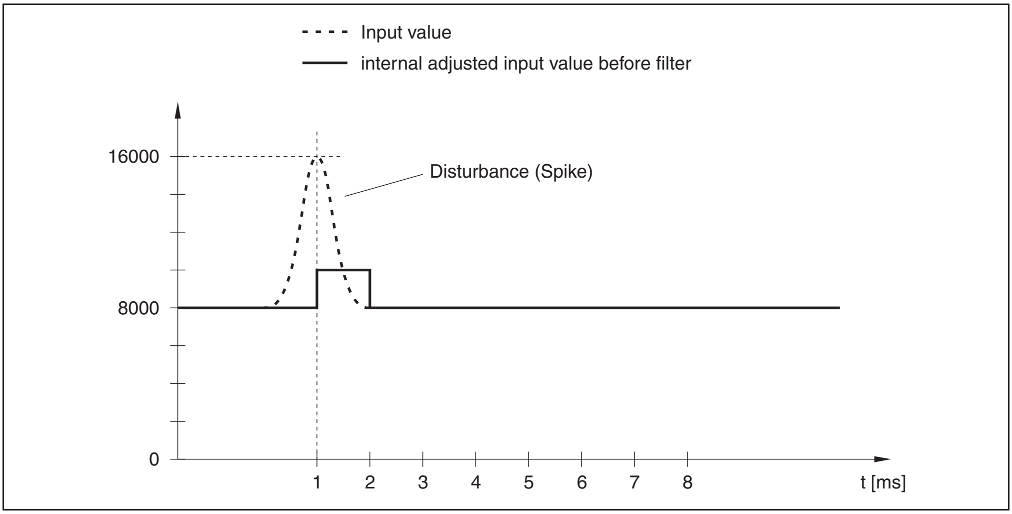
Filter level = 2



Example 2: A disturbance is imposed on the input value. The diagram shows the adjusted input value with the following settings:

Input ramp limitation = 4 = $07FF = 2047

Filter level = 2



Filter level

Depending on the filter level, the input value is more strongly or less strongly evaluated. The evaluation is then made for a possible input ramp limitation.

Formula for the evaluation of the input value:

$$\mathrm{Value}_{new} = \mathrm{Value}_{old} - \frac{\mathrm{Value}_{old}}{\mathrm{FilterLevel}} + \frac{\mathrm{InputValue}}{\mathrm{FilterLevel}}$$

Adjustable filter levels:

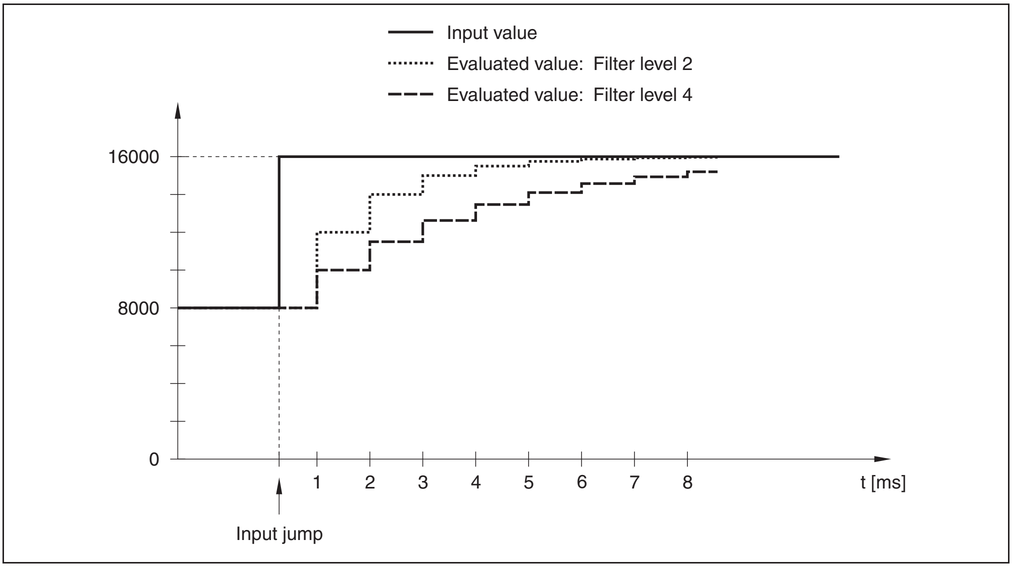
Code	Filter level
0	Filter switched off
1	Filter level 2
2	Filter level 4
3	Filter level 8
4	Filter level 16
5	Filter level 32
6	Filter level 64
7	Filter level 128

The following examples show the function of the input ramp limitation based on an input jump and a disturbance.

Example 1: The input value makes a jump from 8,000 to 16,000. The diagram displays the evaluated value with the following settings:

Input ramp limitation = 0

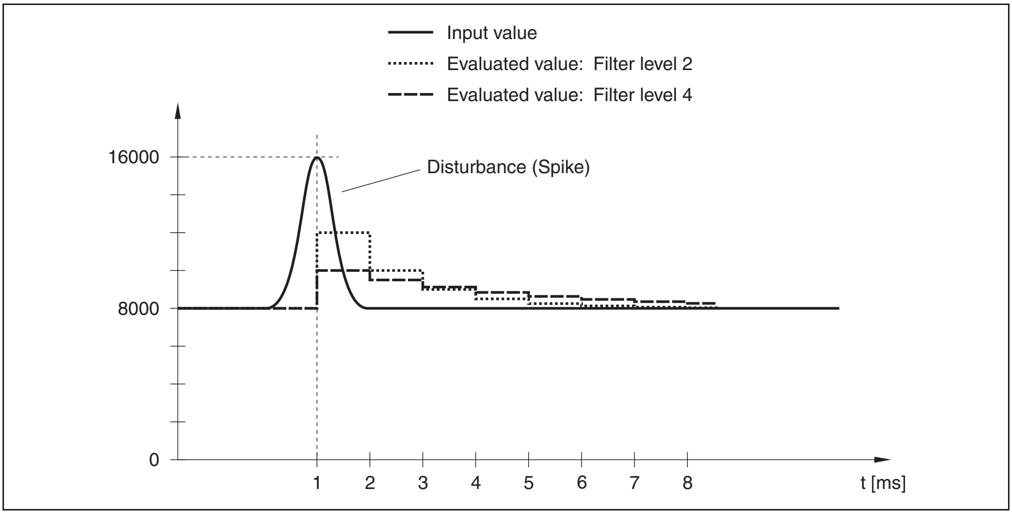
Filter level = 2 or 4



Example 2: A disturbance is imposed on the input value. The diagram shows the evaluated value with the following settings:

Input ramp limitation = 0

Filter level = 2 or 4



10.2 AI2622

10.2.1 General information

The AI2622 module is equipped with two inputs with 12-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

The AI2622 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 analog inputs
- Both current and voltage signals are possible
- 12-bit digital converter resolution

10.2.2 Order data

Model number	Short description	Image
	Analog input module	
X20AI2622	X20 analog input module, 2 inputs, ±10 V / 0 to 20 mA, 12-bit resolution, configurable input filter	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 144: AI2622 - order data

10.2.3 Technical data

Product ID	AI2622	
Short description		
I/O module	2 analog inputs, ±10 V or 0 to 20 mA	
Analog inputs	**Voltage**	**Current**
Input	±10 V or 0 to 20 mA, using different connection terminals	
Input type	Differential input	
Digital converter resolution	±12-bit	12-bit
Conversion time	300 µs for all inputs	
Output format	INT	
Input impedance in signal range	20 MΩ	-
Load	-	<400 Ω
Maximum error at 25 °C Gain Offset	 <0.08 % 1) <0.015 % 2)	 <0.08 % 1) <0.03 % 3)
Input protection	Protection against wiring with supply voltage	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Inputs Channel type	 Yes, with status LED and software status Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 0.8 W	
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C	
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	

Table 145: AI2622 - technical data

Product ID	AI2622
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 145: AI2622 - technical data (cont.)

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.2.4 Additional technical data

Product ID	AI2622	
Analog inputs	**Voltage**	**Current**
Output format	INT \$8001 - \$7FFF / 1 LSB = \$0008 = 2.441 mV	INT \$0000 - \$7FFF / 1 LSB = \$0008 = 4.883 µA
Output of the digital value during overload Above upper limit Below lower limit	 \$7FFF \$8001	 \$7FFF \$0000
Conversion method	SAR	
Input filter	Low pass 3rd order / cut-off frequency 1 kHz	
Max gain drift	<0.006 %/°C [1]	<0.009 %/°C [1]
Max offset drift	<0.002 %/°C [2]	<0.004 %/°C [3]
Common-mode rejection DC 50 Hz	 70 dB 70 dB	
Synchronized zone	±12 V	
Cross-talk between channels	-70 dB	
Non-linearity	<0.025 % [2]	<0.05 % [3]
Isolation voltage betw. channel and bus	500 V_{eff}	
General information		
B&R ID code	\$1B9E	

Table 146: AI2622 - additional technical data

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.2.5 Status LEDs

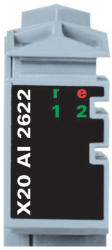
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	Open connection or sensor is disconnected
			Blinking	Overflow or underflow of the input signal
			On	The analog/digital converter is running, value is OK

Table 147: AI2622 - status indicators

10.2.6 Pin assignments

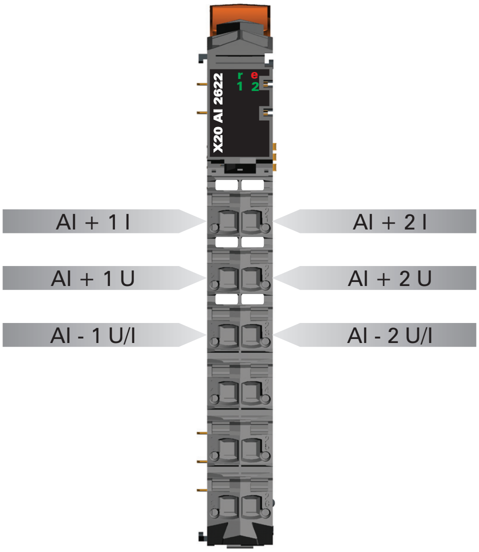


Figure 110: AI2622 - pin assignments

10.2.7 Connection example

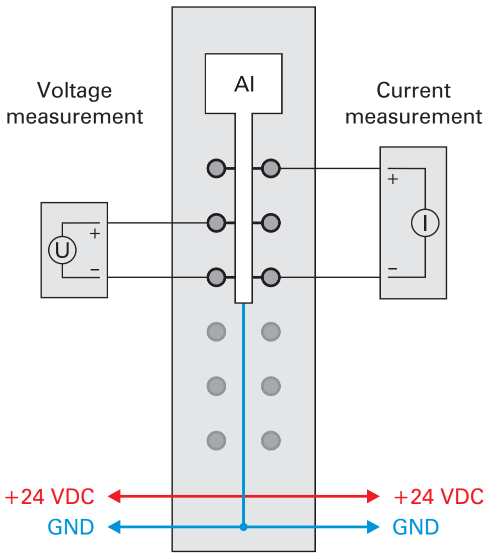


Figure 111: AI2622 - connection example

10.2.8 Input circuit diagram

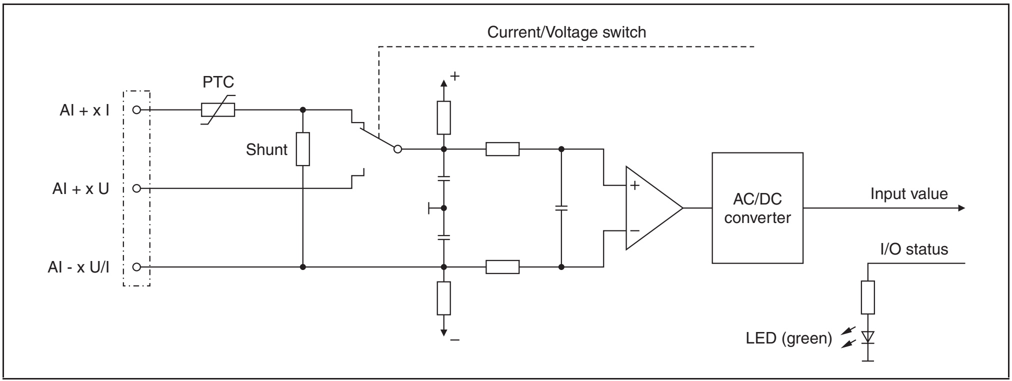


Figure 112: AI2622 - input circuit diagram

10.2.9 Analog inputs

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

10.2.10 Input filter

The module has one input filter and one input ramp limit (see section 10.1.2 "Input filter" on page 258).

10.2.11 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of input signal.

Code	Input signal
0	Voltage signal (default)
1	Current signal

10.2.12 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Voltage signal	Current signal
0	No error	No error
1	Below lower limit value	The input value has a lower limit of $0000. Underflow monitoring is therefore not necessary.
2	Above upper limit value	Above upper limit value
3	Wire break	-

10.2.13 "StatusInput01" register

Bit	Voltage signal description	Current signal description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 1: 00 ... No error 10 ... Above upper limit value
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 2: 00 ... No error 10 ... Above upper limit value
4 - 7	0	0

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Digital value for error
Wire break	+32767 ($7FFF)
Above upper limit value	+32767 ($7FFF)
Below lower limit value	-32767 ($8001)
Invalid value	-32768 ($8000)

10.2.14 B&R ID code

Code for module identification ($1B9E).

10.2.15 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Inputs without filtering	≥100 µs
Inputs with filtering	>500 µs

10.2.16 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Inputs without filtering	300 µs for all inputs
Inputs with filtering	≥1 ms

10.3 AI2632

10.3.1 General information

The AI2632 module is equipped with two inputs with 16-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

The AI2632 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 analog inputs
- Both current and voltage signals are possible
- 16-bit digital converter resolution
- Simultaneous input conversion
- Very fast conversion time

10.3.2 Order data

Model number	Short description	Image
	Analog input module	
X20AI2632	X20 analog input module, 2 inputs, ±10 V / 0 to 20 mA, 16-bit resolution, configurable input filter	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 148: AI2632 - order data

10.3.3 Technical data

Product ID	AI2632	
Short description		
I/O module	2 analog inputs, ±10 V or 0 to 20 mA	
Analog inputs	**Voltage**	**Current**
Input	±10 V or 0 to 20 mA, using different connection terminals	
Input type	Differential input	
Digital converter resolution	±15-bit	15-bit
Conversion time	50 μs for all inputs	
Output format	INT	
Input impedance in signal range	20 MΩ	-
Load	-	<400 Ω
Maximum error at 25 °C Gain Offset	 <0.08 % 1) <0.01 % 2)	 <0.08 % 1) <0.02 % 3)
Input protection	Protection against wiring with supply voltage	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Inputs Channel type	 Yes, with status LED and software status Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.2 W	
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C	
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	

Table 149: AI2632 - technical data

Product ID	AI2632
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 149: AI2632 - technical data (cont.)

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.3.4 Additional technical data

Product ID	AI2632	
Analog inputs	**Voltage**	**Current**
Output format	INT $8001 - $7FFF / 1 LSB = $0001 = 305.176 µV	INT $0000 - $7FFF / 1 LSB = $0001 = 610.352 nA
Output of the digital value during overload Above upper limit Below lower limit	 $7FFF $8001	 $7FFF $0000
Conversion method	SAR	
Input filter	Hardware - low pass 3rd order / cut-off frequency 10 kHz	
Max gain drift	<0.01 %/°C 1)	<0.01 %/°C 1)
Max offset drift	<0.001 %/°C 2)	<0.002 %/°C 3)
Common-mode rejection DC 50 Hz	 70 dB 70 dB	
Synchronized zone	±12 V	
Cross-talk between channels	-70 dB	
Non-linearity	<0.01 % 2)	<0.015 % 3)
Isolation voltage betw. channel and bus	500 V_{eff}	
General information		
B&R ID code	$1BA0	

Table 150: AI2632 - additional technical data

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.3.5 Status LEDs

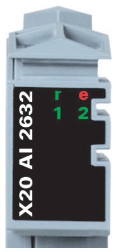
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Double flash	System error: • Violation of the scan time • Synchronization error
	1 - 2	Green	Off	Open connection or sensor is disconnected
			On	The analog/digital converter is running, value is OK

Table 151: AI2632 - status indicators

10.3.6 Pin assignments

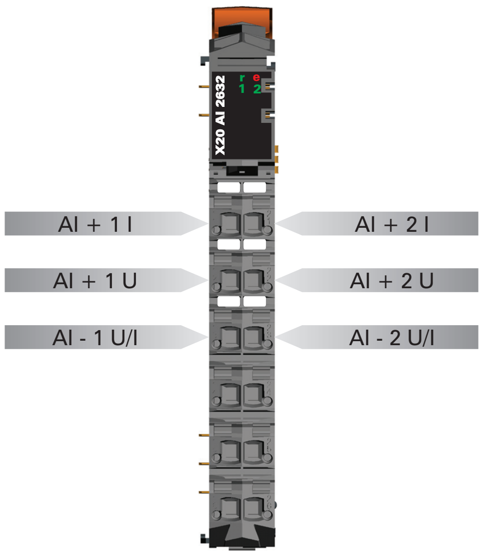


Figure 113: AI2632 - pin assignments

10.3.7 Connection example

To prevent disturbances, the following modules must be separated by at least one module:

- Bus receiver BR9300
- Supply module PS3300
- Supply module PS9400
- CPUs

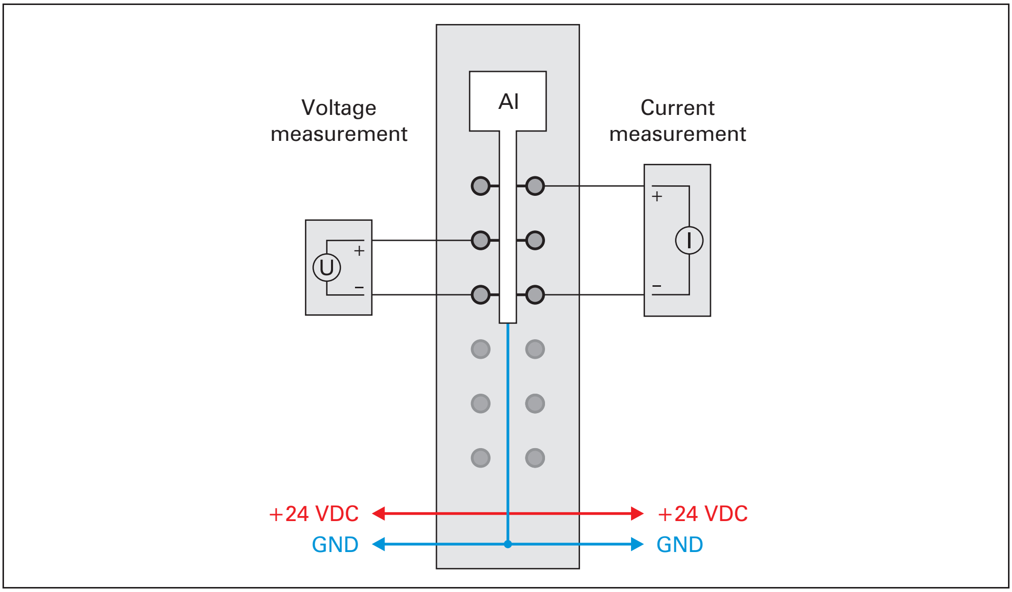


Figure 114: AI2632 - connection example

10.3.8 Input circuit diagram

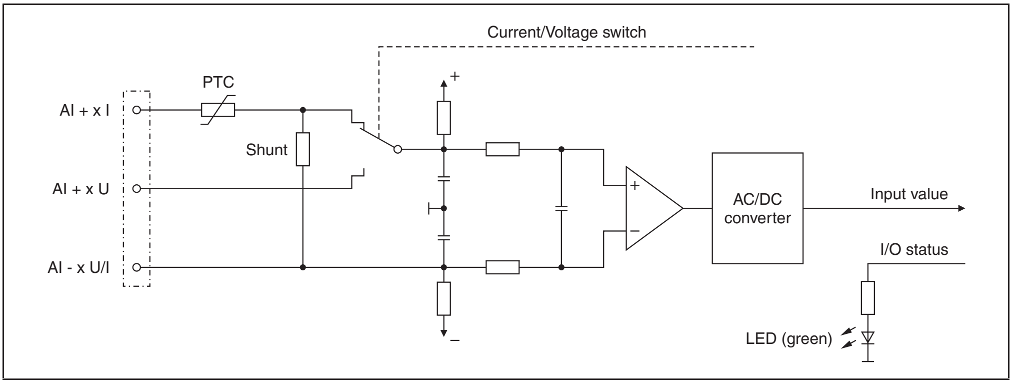


Figure 115: AI2632 - input circuit diagram

10.4 AI4622

10.4.1 General information

The AI4622 module is equipped with four inputs with 12-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

- 4 analog inputs
- Both current and voltage signals are possible
- 12-bit digital converter resolution

10.4.2 Order data

Model number	Short description	Image
	Analog input module	
X20AI4622	X20 analog input module, 4 inputs, ±10 V / 0 to 20 mA, 12-bit resolution, configurable input filter	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 152: AI4622 - order data

10.4.3 Technical data

Product ID	AI4622	
Short description		
I/O module	4 analog inputs, ±10 V or 0 to 20 mA	
Analog inputs	**Voltage**	**Current**
Input	±10 V or 0 to 20 mA, using different connection terminals	
Input type	Differential input	
Digital converter resolution	±12-bit	12-bit
Conversion time	400 µs for all inputs	
Output format	INT	
Input impedance in signal range	20 MΩ	-
Load	-	<400 Ω
Maximum error at 25 °C Gain Offset	 <0.08 % 1) <0.015 % 2)	 <0.08 % 1) <0.03 % 3)
Input protection	Protection against wiring with supply voltage	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Inputs Channel type	 Yes, with status LED and software status Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.1 W	
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C	
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	

Table 153: AI4622 - technical data

Product ID	AI4622
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 153: AI4622 - technical data (cont.)

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.4.4 Additional technical data

Product ID	AI4622	
Analog inputs	**Voltage**	**Current**
Output format	INT $8001 - $7FFF / 1 LSB = $0008 = 2.441 mV	INT $0000 - $7FFF / 1 LSB = $0008 = 4.883 µA
Output of the digital value during overload Above upper limit Below lower limit	 $7FFF $8001	 $7FFF $0000
Conversion method	SAR	
Input filter	Low pass 3rd order / cut-off frequency 1 kHz	
Max gain drift	<0.006 %/°C 1)	<0.009 %/°C 1)
Max offset drift	<0.002 %/°C 2)	<0.004 %/°C 3)
Common-mode rejection DC 50 Hz	 70 dB 70 dB	
Synchronized zone	±12 V	
Cross-talk between channels	-70 dB	
Non-linearity	<0.025 % 2)	<0.05 % 3)
Isolation voltage betw. channel and bus	500 V_{eff}	
General information		
B&R ID code	$1BAA	

Table 154: AI4622 - additional technical data

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.4.5 Status LEDs

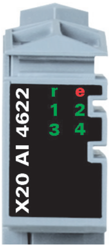
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Green	Off	Open connection or sensor is disconnected
			Blinking	Overflow or underflow of the input signal
			On	The analog/digital converter is running, value is OK

Table 155: AI4622 - status indicators

10.4.6 Pin assignments

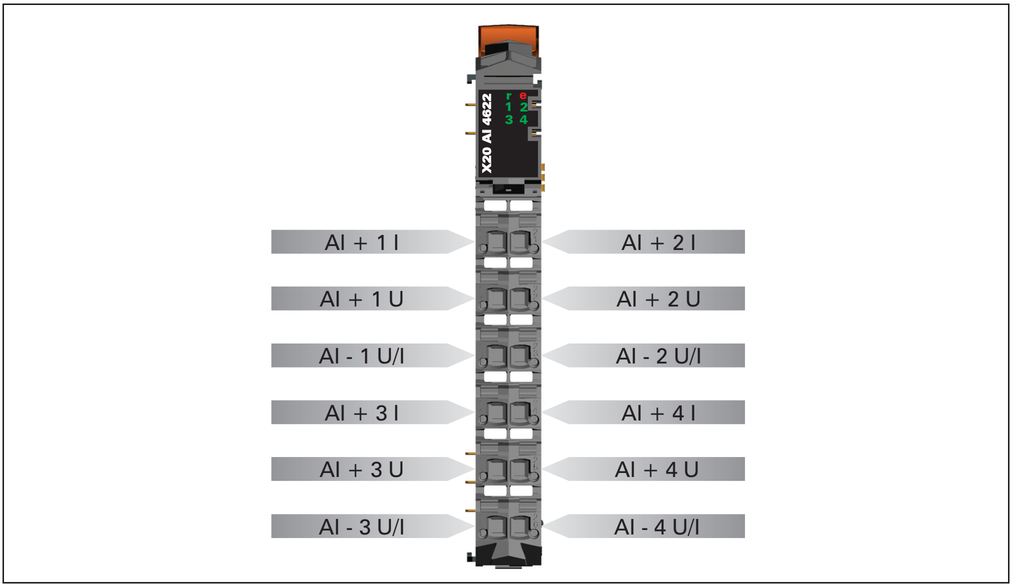


Figure 116: AI4622 - pin assignments

10.4.7 Connection example

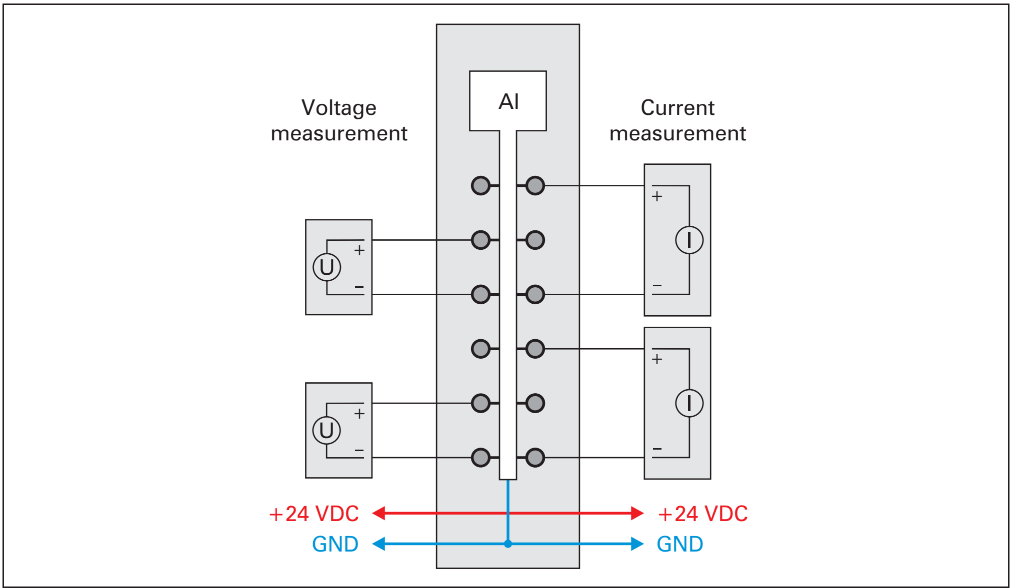


Figure 117: AI4622 - connection example

10.4.8 Input circuit diagram

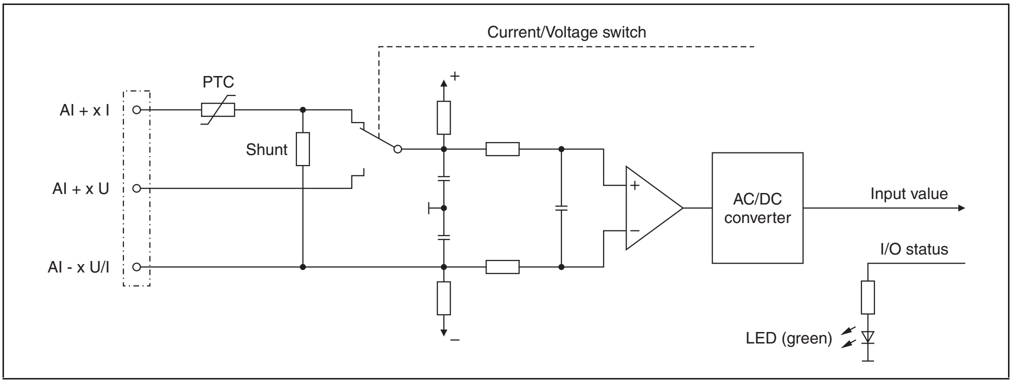


Figure 118: AI4622 - input circuit diagram

10.4.9 Analog inputs

The input status is registered with a fixed offset with respect to the network cycle and is transferred in the same cycle.

10.4.10 Input filter

The module has one input filter and one input ramp limit (see section 10.1.2 "Input filter" on page 258).

10.4.11 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of input signal.

Code	Input signal
0	Voltage signal (default)
1	Current signal

10.4.12 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Voltage signal	Current signal
0	No error	No error
1	Below lower limit value	The input value has a lower limit of $0000. Underflow monitoring is therefore not necessary.
2	Above upper limit value	Above upper limit value
3	Wire break	-

10.4.13 "StatusInput01" register

Bit	Voltage signal description	Current signal description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 1: 00 ... No error 10 ... Above upper limit value
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 2: 00 ... No error 10 ... Above upper limit value
4 - 5	Channel 3: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 3: 00 ... No error 10 ... Above upper limit value
6 - 7	Channel 4: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break	Channel 4: 00 ... No error 10 ... Above upper limit value

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Digital value for error
Wire break	+32767 ($7FFF)
Above upper limit value	+32767 ($7FFF)
Below lower limit value	-32767 ($8001)
Invalid value	-32768 ($8000)

10.4.14 B&R ID code

Code for module identification ($1BAA).

10.4.15 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Inputs without filtering	≥100 μs
Inputs with filtering	>500 μs

10.4.16 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Inputs without filtering	300 μs for all inputs
Inputs with filtering	≥1 ms

10.5 AI4632

10.5.1 General information

The AI4632 module is equipped with four inputs with 16-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

- 4 analog inputs
- Both current and voltage signals are possible
- 16-bit digital converter resolution
- Simultaneous input conversion
- Very fast conversion time

10.5.2 Order data

Model number	Short description	Image
	Analog input module	
X20AI4632	X20 analog input module, 4 inputs, ±10 V / 0 to 20 mA, 16-bit resolution, configurable input filter	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 156: AI4632 - order data

10.5.3 Technical data

Product ID	AI4632	
Short description		
I/O module	4 analog inputs, ±10 V or 0 to 20 mA	
Analog inputs	**Voltage**	**Current**
Input	±10 V or 0 to 20 mA, using different connection terminals	
Input type	Differential input	
Digital converter resolution	±15-bit	15-bit
Conversion time	50 µs for all inputs	
Output format	INT	
Input impedance in signal range	20 MΩ	-
Load	-	<400 Ω
Maximum error at 25 °C Gain Offset	 <0.08 % 1) <0.01 % 2)	 <0.08 % 1) <0.02 % 3)
Input protection	Protection against wiring with supply voltage	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Inputs Channel type	 Yes, with status LED and software status Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.5 W	
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C	
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	

Table 157: AI4632 - technical data

Product ID	AI4632
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 157: AI4632 - technical data (cont.)

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.5.4 Additional technical data

Product ID	AI4632	
Analog inputs	**Voltage**	**Current**
Output format	INT $8001 - $7FFF / 1 LSB = $0001 = 305.176 µV	INT $0000 - $7FFF / 1 LSB = $0001 = 610.352 nA
Output of the digital value during overload Above upper limit Below lower limit	 $7FFF $8001	 $7FFF $0000
Conversion method	SAR	
Input filter	Hardware - low pass 3rd order / cut-off frequency 10 kHz	
Max gain drift	<0.01 %/°C 1)	<0.01 %/°C 1)
Max offset drift	<0.001 %/°C 2)	<0.002 %/°C 3)
Common-mode rejection DC 50 Hz	 70 dB 70 dB	
Synchronized zone	±12 V	
Cross-talk between channels	-70 dB	
Non-linearity	<0.01 % 2)	<0.015 % 3)
Isolation voltage betw. channel and bus	500 V_{eff}	
General information		
B&R ID code	$1BA1	

Table 158: AI4632 - additional technical data

1) Based on the current measurement value.
2) Refers to the 20 V measurement range.
3) Refers to the 20 mA measurement range.

10.5.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Double flash	System error: • Violation of the scan time • Synchronization error
	1 - 4	Green	Off	Open connection or sensor is disconnected
			On	The analog/digital converter is running, value is OK

Table 159: AI4632 - status indicators

10.5.6 Pin assignments

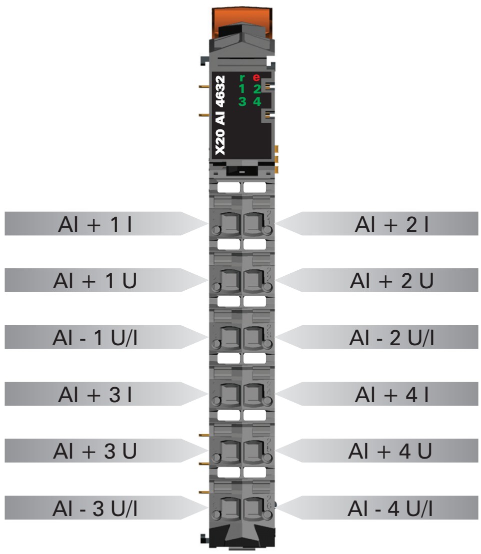


Figure 119: AI4632 - pin assignments

10.5.7 Connection example

To prevent disturbances, the following modules must be separated by at least one module:

- Bus receiver BR9300
- Supply module PS3300
- Supply module PS9400
- CPUs

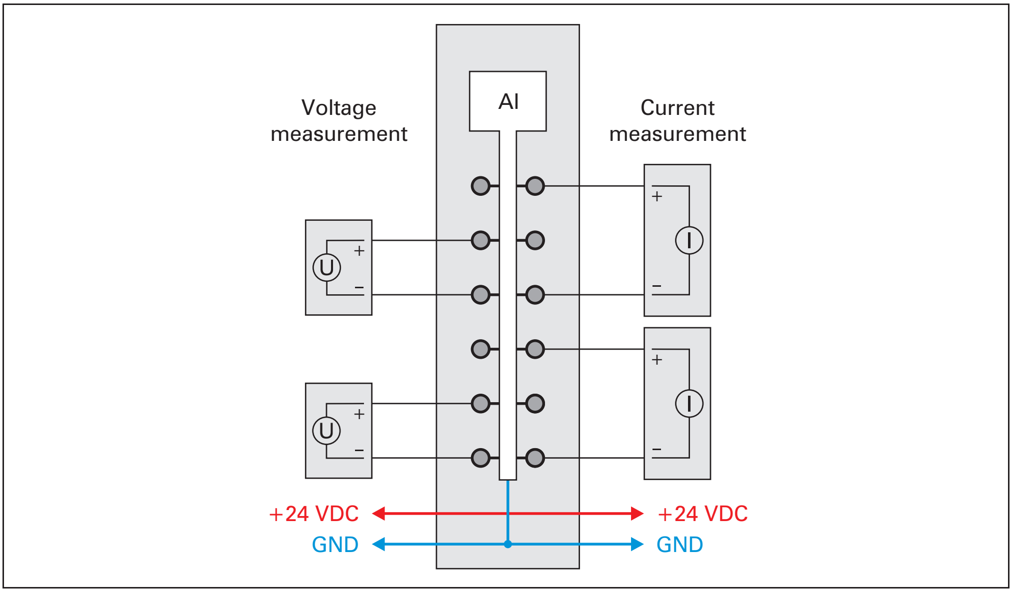


Figure 120: AI4632 - connection example

10.5.8 Input circuit diagram

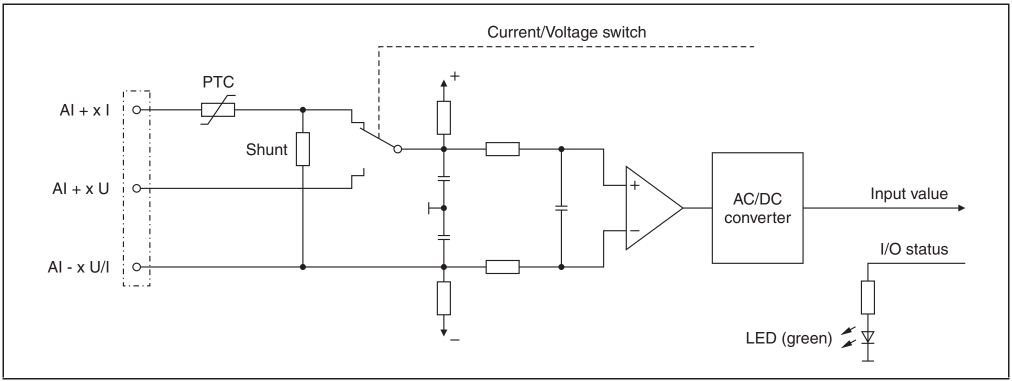


Figure 121: AI4632 - input circuit diagram

11. Analog output modules

11.1 General information

Analog output modules convert PLC internal numerical values into voltages or currents. The numbers to be converted must be in 16-bit 2s complement. The conversion takes place independent of the resolution of the output module used.

Every channel on an analog output module has a status LED.

11.1.1 Overview

Module	AO2622	AO2632	AO4622	AO4632
Number of outputs	2	2	4	4
Output signal, rated	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]	±10 V or 0 to 20 mA [1)]
Digital converter resolution	12-bit	16-bit	12-bit	16-bit

Table 160: Overview of analog output modules

1) Using different connection terminals.

11.2 AO2622

11.2.1 General information

The AO2622 module is equipped with two outputs with 12-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

The AO2622 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 analog outputs
- Both current and voltage signals are possible
- 12-bit digital converter resolution

11.2.2 Order data

Model number	Short description	Image
	Analog output module	
X20AO2622	X20 analog output module, 2 outputs, ±10 V / 0 to 20 mA, 12-bit resolution	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 161: AO2622 - order data

11.2.3 Technical data

Product ID	AO2622
Short description	
I/O module	2 analog outputs, ±10 V or 0 to 20 mA
Analog outputs	
Output	±10 V or 0 to 20 mA, using different connection terminals
Digital converter resolution	12-bit
Conversion time	200 µs for all outputs
Power on/off behavior	Internal enable relay for boot procedure and error
Maximum error at 25 °C Gain Offset	 <0.15 %, based on the current output value <0.05 %, based on the entire output range
Output protection	Short circuit protection
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Channel type	 Yes, with status LED and software status Yes, with software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.1 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 162: AO2622 - technical data

11.2.4 Additional technical data

Product ID	AO2622
Analog outputs	
Output format Voltage Current	 INT \$8001 - \$7FFF / 1 LSB = \$0010 = 4.882 mV INT \$0000 - \$7FFF / 1 LSB = \$0010 = 9.766 µA
Load per channel Voltage Current	 Max. ±10 mA (load ≥ 1 kΩ) Max load 600 Ω
Short circuit protection	Current limitation ±40 mA
Output filter	Low pass 1st order / cut-off frequency 10 kHz
Maximum gain drift	<0.02 % per °C, based on the current output value
Maximum offset drift	<0.032 % per °C, based on the entire output range
Error caused by load change Voltage Current	 Max. 0.02 % (from 10 MΩ → 1 kΩ, resistive) Max. 0.5 % (from 1 Ω → 600 Ω, resistive)
Response time for output changes over entire range	1 ms
Non-linearity	<0.007 %, based on the output range
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	\$1BA2

Table 163: AO2622 - additional technical data

11.2.5 Status LEDs

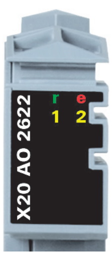
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Orange	Off	Value = 0
			On	Value ≠ 0

Table 164: AO2622 - status indicators

11.2.6 Pin assignments

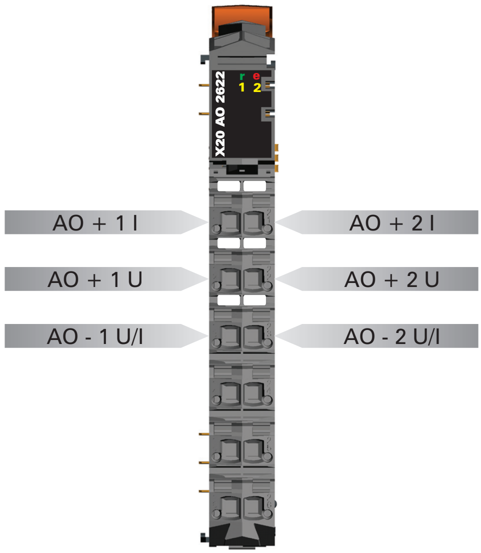


Figure 122: AO2622 - pin assignments

11.2.7 Connection example

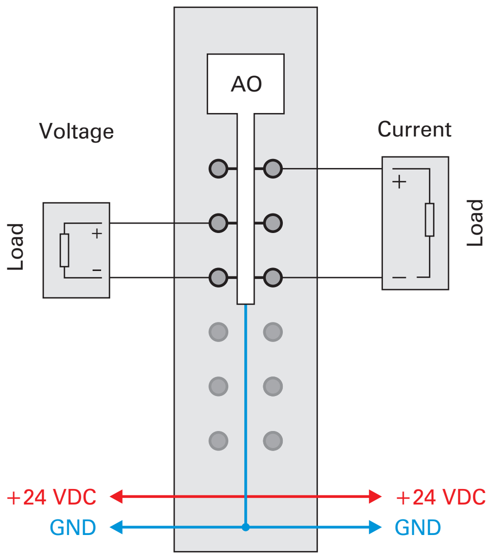


Figure 123: AO2622 - connection example

11.2.8 Output circuit diagram

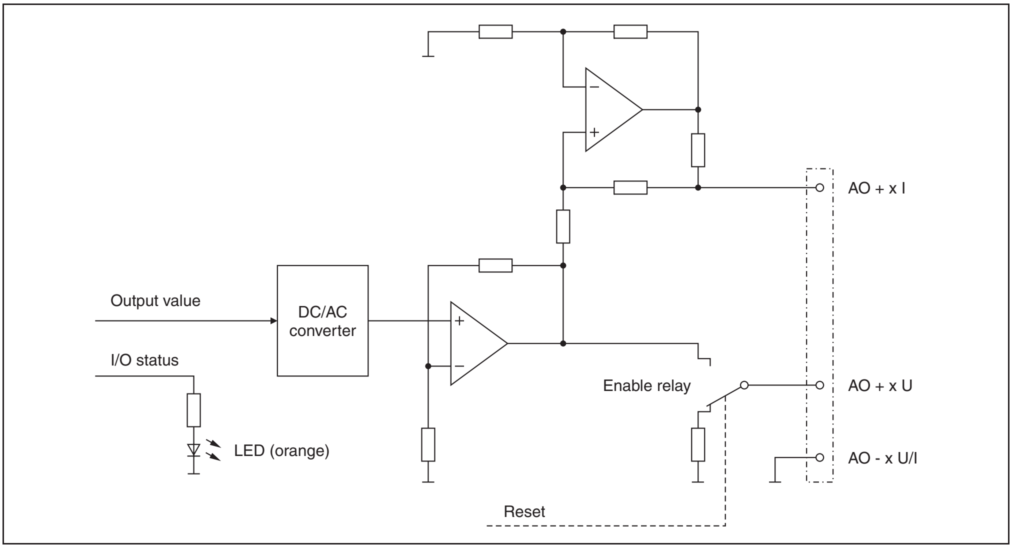


Figure 124: AO2622 - output circuit diagram

11.2.9 Analog outputs

Each channel can be configured for either current or voltage signals. The type of signal is also determined by the connection terminals used.

11.2.10 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of output signal.

Code	Output signal
0	Voltage signal (default)
1	Current signal

11.2.11 B&R ID code

Code for module identification ($1BA2).

11.2.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Minimum cycle time	≥250 µs

11.2.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
For all outputs	<300 µs

11.3 AO2632

11.3.1 General information

The AO2632 module is equipped with two outputs with 16-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

The AO2632 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 analog outputs
- Both current and voltage signals are possible
- 16-bit digital converter resolution

11.3.2 Order data

Model number	Short description	Image
	Analog output module	
X20AO2632	X20 analog output module, 2 outputs, ±10 V / 0 to 20 mA, 16-bit resolution	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 165: AO2632 - order data

11.3.3 Technical data

Product ID	AO2632
Short description	
I/O module	2 analog outputs, ±10 V or 0 to 20 mA
Analog outputs	
Output	±10 V or 0 to 20 mA, using different connection terminals
Digital converter resolution	16-bit

Table 166: AO2632 - technical data

Product ID	AO2632	
Conversion time	50 µs for all outputs	
Power on/off behavior	Internal enable relay for boot procedure and error	
Maximum error at 25 °C Gain Offset	 <0.045 %, based on the current output value <0.025 %, based on the entire output range	
Output protection	Short circuit protection	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Channel type	 Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	Rev. <B0 Typ. 0.01 W Typ. 1.6 W	Rev. ≥B0 Typ. 0.01 W Typ. 1.2 W
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	Rev. <B0 0 °C to +50 °C 0 °C to +45 °C	Rev. ≥B0 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	
Mechanical characteristics		
Spacing	12.5 +0.2 mm	
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately	

Table 166: AO2632 - technical data (cont.)

11.3.4 Additional technical data

Product ID	AO2632
Analog outputs	
Output format Voltage Current	 INT \$8001 - \$7FFF / 1 LSB = \$0001 = 305.176 µV INT \$0000 - \$7FFF / 1 LSB = \$0001 = 610.352 nA
Load per channel Voltage Current	 Max. ±10 mA (load ≥ 1 kΩ) Max load 600 Ω
Short circuit protection	Current limitation ±40 mA
Output filter	Low pass 1st order / cut-off frequency 10 kHz
Maximum gain drift	<0.015 % per °C, based on the current output value
Maximum offset drift	<0.013 % per °C, based on the entire output range
Error caused by load change Voltage Current	 Max. 0.02 % (from 10 MΩ → 1 kΩ, resistive) Max. 0.5 % (from 1 Ω → 600 Ω, resistive)
Response time for output changes over entire range	1 ms
Non-linearity	<0.007 %, based on the output range
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	\$1BA4

Table 167: AO2632 - additional technical data

11.3.5 Status LEDs

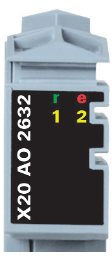
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	1 - 2	Orange	Off	Value = 0
			On	Value ≠ 0

Table 168: AO2632 - status indicators

11.3.6 Pin assignments

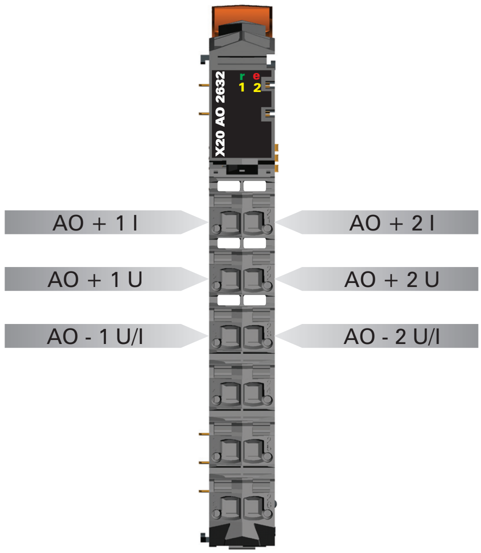


Figure 125: AO2632 - pin assignments

11.3.7 Connection example

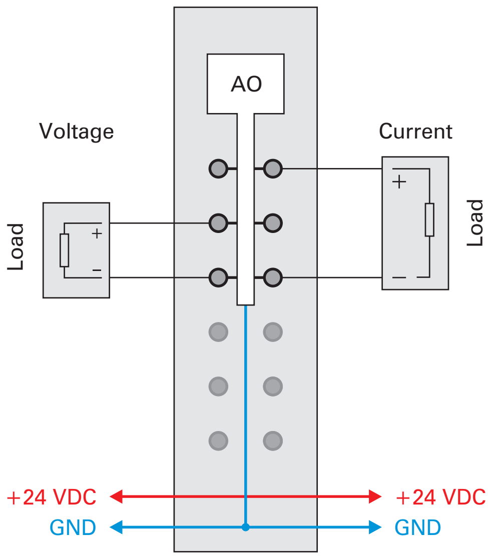


Figure 126: AO2632 - connection example

11.3.8 Output circuit diagram

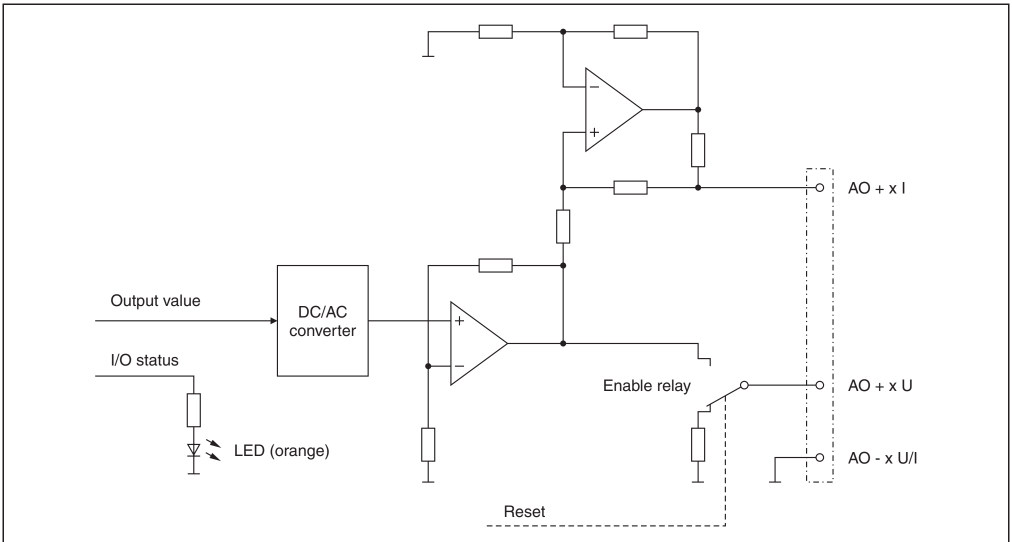


Figure 127: AO2632 - output circuit diagram

11.3.9 Analog outputs

Each channel can be configured for either current or voltage signals. The type of signal is also determined by the connection terminals used.

11.3.10 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of output signal.

Code	Output signal
0	Voltage signal (default)
1	Current signal

11.3.11 B&R ID code

Code for module identification ($1BA4).

11.3.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Minimum cycle time	≥200 µs

11.3.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
For all outputs	≥200 µs

11.4 AO4622

11.4.1 General information

The AO4622 module is equipped with four outputs with 12-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

- 4 analog outputs
- Both current and voltage signals are possible
- 12-bit digital converter resolution

11.4.2 Order data

Model number	Short description	Image
	Analog output module	
X20AO4622	X20 analog output module, 4 outputs, ±10 V / 0 to 20 mA, 12-bit resolution	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 169: AO4622 - order data

11.4.3 Technical data

Product ID	AO4622
Short description	
I/O module	4 analog outputs, ±10 V or 0 to 20 mA
Analog outputs	
Output	±10 V or 0 to 20 mA, using different connection terminals
Digital converter resolution	12-bit
Conversion time	300 µs for all outputs
Power on/off behavior	Internal enable relay for boot procedure and error
Maximum error at 25 °C Gain Offset	 <0.080 %, based on the current output value <0.050 %, based on the entire output range
Output protection	Short circuit protection
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Channel type	 Yes, with status LED and software status Yes, with software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 170: AO4622 - technical data

11.4.4 Additional technical data

Product ID	AO4622
Analog outputs	
Output format Voltage Current	 INT \$8001 - \$7FFF / 1 LSB = \$0010 = 4.882 mV INT \$0000 - \$7FFF / 1 LSB = \$0010 = 9.766 µA
Load per channel Voltage Current	 Max. ±10 mA (load ≥ 1 kΩ) Max load 600 Ω
Short circuit protection	Current limitation ±40 mA
Output filter	Low pass 1st order / cut-off frequency 10 kHz
Maximum gain drift	<0.015 % per °C, based on the current output value
Maximum offset drift	<0.032 % per °C, based on the entire output range
Error caused by load change Voltage Current	 Max. 0.02 % (from 10 MΩ → 1 kΩ, resistive) Max. 0.5 % (from 1 Ω → 600 Ω, resistive)
Response time for output changes over entire range	500 µs
Non-linearity	<0.005 %, based on the output range
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	\$1BA3

Table 171: AO4622 - additional technical data

11.4.5 Status LEDs

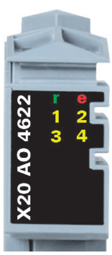
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Orange	Off	Value = 0
			On	Value ≠ 0

Table 172: AO4622 - status indicators

11.4.6 Pin assignments

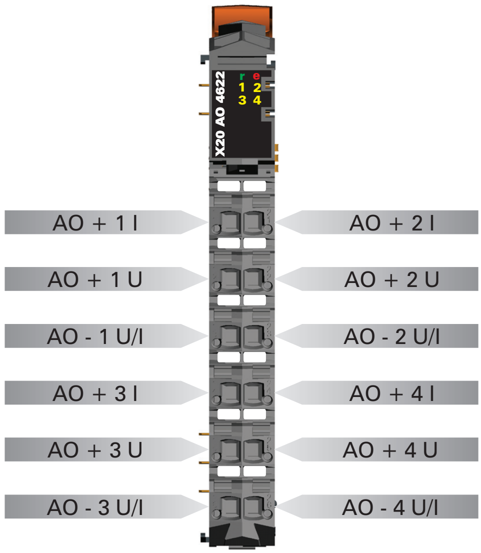


Figure 128: AO4622 - pin assignments

11.4.7 Connection example

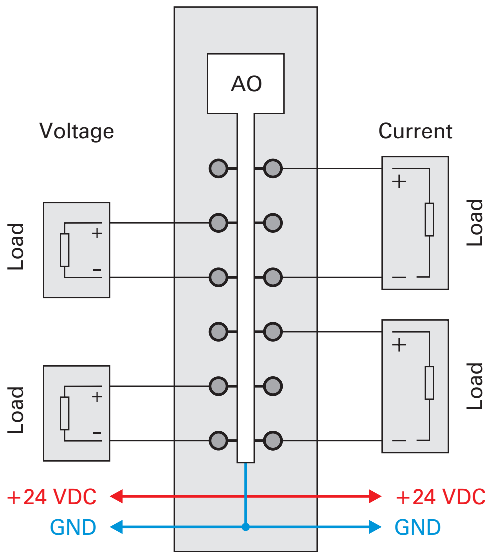


Figure 129: AO4622 - connection example

11.4.8 Output circuit diagram

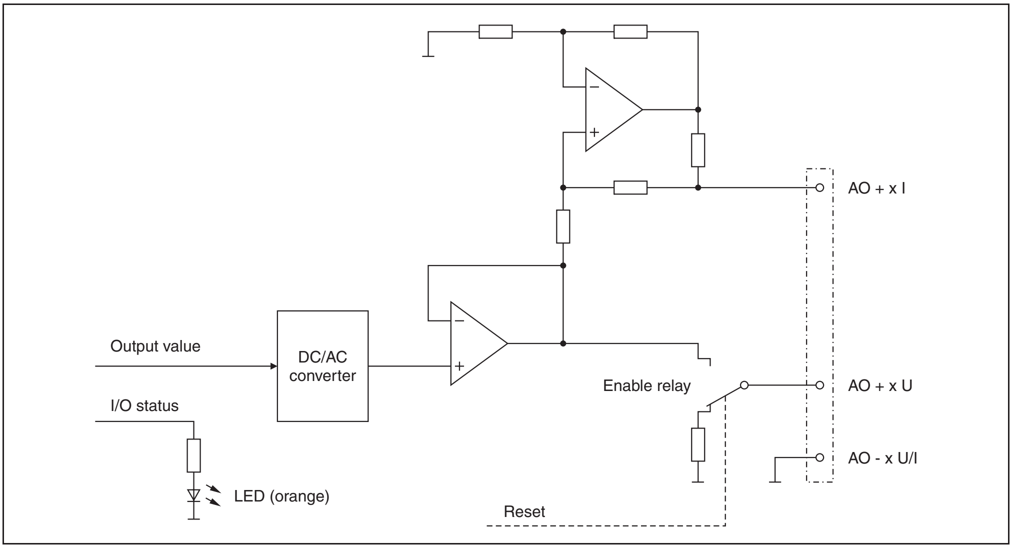


Figure 130: AO4622 - output circuit diagram

11.4.9 Analog outputs

Each channel can be configured for either current or voltage signals. The type of signal is also determined by the connection terminals used.

11.4.10 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of output signal.

Code	Output signal
0	Voltage signal (default)
1	Current signal

11.4.11 B&R ID code

Code for module identification ($1BA3).

11.4.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Minimum cycle time	≥250 µs

11.4.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
For all outputs	<400 µs

11.5 AO4632

11.5.1 General information

The AO4632 module is equipped with four outputs with 16-bit digital converter resolution. Using different connection terminals you can select between the current and voltage signal.

- 4 analog outputs
- Both current and voltage signals are possible
- 16-bit digital converter resolution

11.5.2 Order data

Model number	Short description	Image
	Analog output module	
X20AO4632	X20 analog output module, 4 outputs, ±10 V / 0 to 20 mA, 16-bit resolution	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 173: AO4632 - order data

11.5.3 Technical data

Product ID	AO4632
Short description	
I/O module	4 analog outputs, ±10 V or 0 to 20 mA
Analog outputs	
Output	±10 V or 0 to 20 mA, using different connection terminals
Digital converter resolution	16-bit
Conversion time	50 µs for all outputs
Power on/off behavior	Internal enable relay for boot procedure and error

Table 174: AO4632 - technical data

Product ID	AO4632	
Maximum error at 25 °C Gain Offset	 <0.040 %, based on the current output value <0.022 %, based on the entire output range	
Output protection	Short circuit protection	
General information		
Status indicators	I/O function per channel, operating state, module status	
Diagnostics Module run error Channel type	 Yes, with status LED and software status Yes, with software status	
Electrical isolation Channel - Bus Channel - Channel	 Yes No	
Power consumption Bus I/O internal	Rev. <B0 Typ. 0.01 W Typ. 2.0 W	Rev. ≥B0 Typ. 0.01 W Typ. 1.6 W
Certification	CE, C-UL-US, GOST-R (in development)	
Operational conditions		
Operating temperature Horizontal installation Vertical installation	Rev. <B0 0 °C to +45 °C 0 °C to +40 °C	Rev. ≥B0 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing	
Mounting orientation	Horizontal or vertical	
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m	
Protection type	IP20	
Storage and transport conditions		
Temperature	-25 °C to +70 °C	
Humidity	5 to 95%, non-condensing	
Mechanical characteristics		
Spacing	12.5 +0.2 mm	
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately	

Table 174: AO4632 - technical data (cont.)

11.5.4 Additional technical data

Product ID	AO4632
Analog outputs	
Output format Voltage Current	 INT $8001 - $7FFF / 1 LSB = $0001 = 305.176 µV INT $0000 - $7FFF / 1 LSB = $0001 = 610.352 nA
Load per channel Voltage Current	 Max. ±10 mA (load ≥ 1 kΩ) Max load 600 Ω
Short circuit protection	Current limitation ±40 mA
Output filter	Low pass 1st order / cut-off frequency 10 kHz
Maximum gain drift	<0.01 % per °C, based on the current output value
Maximum offset drift	<0.012 % per °C, based on the entire output range
Error caused by load change Voltage Current	 Max. 0.02 % (from 10 MΩ → 1 kΩ, resistive) Max. 0.5 % (from 1 Ω → 600 Ω, resistive)
Response time for output changes over entire range	500 µs
Non-linearity	<0.005 %, based on the output range
Isolation voltage betw. channel and bus	500 V_{eff}
General information	
B&R ID code	$1BA5

Table 175: AO4632 - additional technical data

11.5.5 Status LEDs

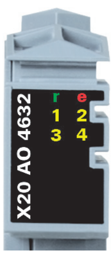
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	1 - 4	Orange	Off	Value = 0
			On	Value ≠ 0

Table 176: AO4632 - status indicators

11.5.6 Pin assignments

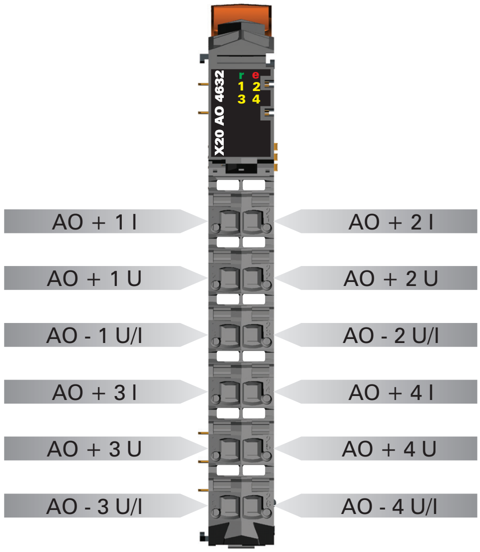


Figure 131: AO4632 - pin assignments

11.5.7 Connection example

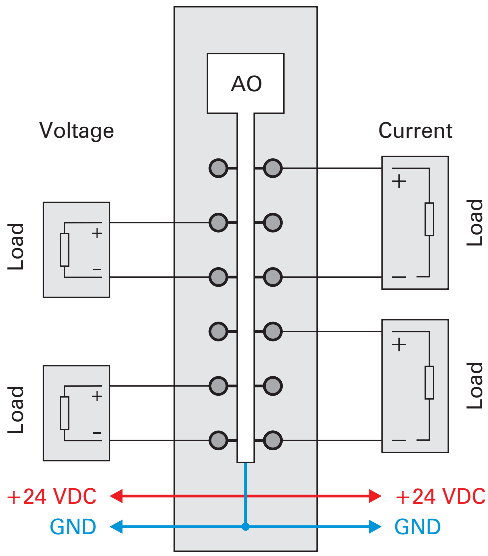


Figure 132: AO4632 - connection example

11.5.8 Output circuit diagram

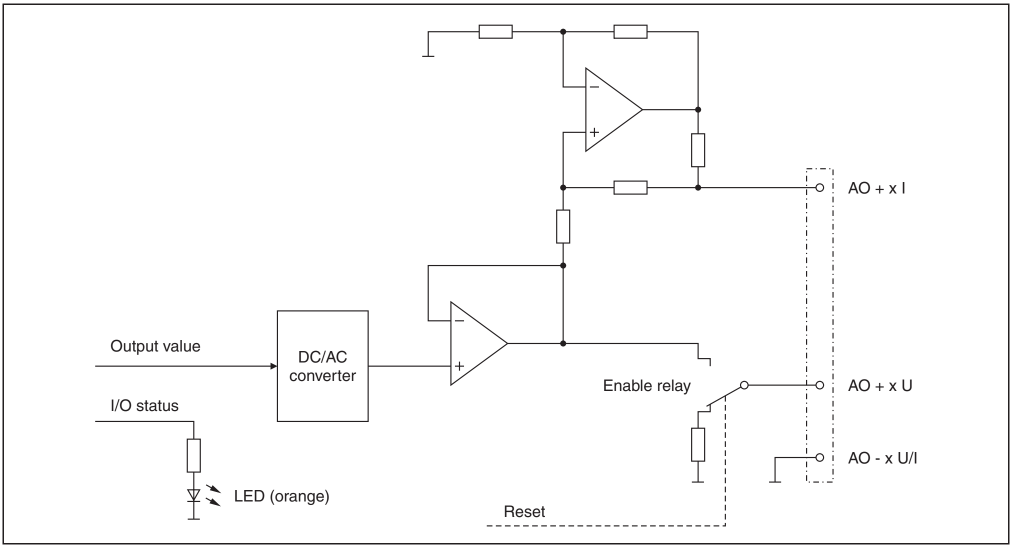


Figure 133: AO4632 - output circuit diagram

11.5.9 Analog outputs

Each channel can be configured for either current or voltage signals. The type of signal is also determined by the connection terminals used.

11.5.10 Channel type

Each channel is capable of either current or voltage signals.

The type of signal is determined by the connection terminals used. Since current and voltage require different adjustment values, it is also necessary to configure the desired type of output signal.

Code	Output signal
0	Voltage signal (default)
1	Current signal

11.5.11 B&R ID code

Code for module identification ($1BA5).

11.5.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Minimum cycle time	≥200 µs

11.5.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
For all outputs	≥200 µs

12. Temperature modules

12.1 General information

Temperature measurement values are converted into number values which can be processed by the PLC using temperature modules.

In the PLC, the number values are always in 16-bit, 2s complement regardless of the resolution. In this way, the resolution (number of steps) of the temperature module does not have to be considered when creating the application program.

For temperature measurements, the temperature module returns the measured value in 0.1° steps. That means, a result of 750 corresponds to 75.0° C. The data format 0.1° C is supported as standard by all temperature modules.

Every channel on temperature module module has a status LED.

12.1.1 Overview

Module	AT2222	AT4222	AT2402	AT6402
Number of channels	2	4	2	6
Measurement area	-200 to +850 °C	-200 to +850 °C	-270 to +1768 °C	-270 to +1768 °C
Sensor	PT100 PT1000	PT100 PT1000	FeCuNi, Type J NiCrNi, Type K PtRhPt, Type S	FeCuNi, Type J NiCrNi, Type K PtRhPt, Type S
Resistance measurement	Yes	Yes	-	-
Raw value evaluation	-	-	Yes	Yes
Digital converter resolution	16-bit	16-bit	16-bit	16-bit

Table 177: Overview of temperature modules

12.2 AT2222

12.2.1 General information

The AT2222 module is equipped with two inputs for PT100/PT1000 resistance temperature measurement. The AT2222 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 inputs for resistance temperature measurement
- For PT100 and PT1000
- Sensor type can be be set for each channel
- Direct resistance measurement as well
- 2 or 3 wire connection can be configured for each module
- Filter time can be configured

12.2.2 Order data

Model number	Short description	Image
	Temperature module	
X20AT2222	X20 temperature input module, 2 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 178: AT2222 - order data

12.2.3 Technical data

Product ID	AT2222
Short description	
I/O module	2 inputs for PT100 or PT1000 resistance temperature measurement
Temperature inputs - resistance measurement	
Input	Resistance measurement with constant current supply for 2 or 3-wire connection
Digital converter resolution	16-bit
Filter time	Configurable between 1 ms and 66.7 ms
Conversion time 1 channel 2 channels	 20 ms at 50 Hz filter 80 ms at 50 Hz filter
Output format	INT or UINT for resistance measurement
Maximum error at 25 °C Gain Offset	 <0.037 %, based on the current resistance value <0.0015 %, based on the entire resistance range
Sensor PT100 PT1000	Can be set per channel -200 °C to +850 °C -200 °C to +850 °C
Resistance measurement range	0.1 Ω to 4500 Ω / 0.05 Ω to 2250 Ω
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Inputs	 Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	 Yes No
Power consumption Bus I/O internal	 Typ. 0.01 W Typ. 1.1 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20

Table 179: AT2222 - technical data